Article

Unraveling Noise in 3D Point Clouds: A Comprehensive Survey

Alejandro Garnung Menéndez [1]

[1] Department of Electrical, Computer Electronics and Systems Engineering, University of Oviedo, 33204, Oviedo, Spain; uo269564@uniovi.es (A.G.); garnungalejandro@gmail.com (A.G.)
* Correspondence: Alejandro Garnung Menéndez; uo269564@uniovi.es; garnungalejandro@gmail.com

Abstract: Noise contamination in three-dimensional data remains a persistent challenge in reconstruction pipelines, with critical implications for emerging technologies such as autonomous navigation, cultural heritage preservation, and industrial inspection. This work presents an innovative three-axis taxonomy that unifies the classification and mitigation of noise in 3D point clouds. Our framework systematically categorizes noise sources into: (1) statistical models, (2) scene interferences, and (3) reconstruction imperfections. Through critical analysis, we reveal fundamental limitations, practical challenges, and adaptive strategies employed in specific scenarios. Among the unresolved challenges are real-time processing of dynamic scenes, separation of overlapping noise types, and biases in learning models toward specific sensor geometries. This research consolidates fragmented knowledge into actionable guidelines, equipping professionals to design robust 3D systems and guiding researchers toward impactful innovations in noise-adaptive sensing.

Keywords: 3D denoising; point cloud processing; noise taxonomy; statistical noise models.

---

1. Introduction

Noise in three-dimensional data presents a complex and multifaceted challenge that significantly impacts reconstruction pipelines and downstream tasks such as object interpretation and scene analysis. Point clouds and other 3D representations, including voxels, meshes, and depth maps (e.g., RGB-D, ToF. etc.), are acquired from real-world environments that are inherently imperfect. These data are often corrupted by noise, outliers, occlusions, and clutter, which introduce irregular shadows and incomplete observations. Furthermore, the natural variability of objects, influenced by factors such as viewpoint, illumination, and positioning [1], adds additional complexity to noise characterization and mitigation.

Point clouds are considered the canonical representation of 3D data, as they provide a universal format for translating and unifying all other 3D representations, including raw sensor data, multiple images, depth maps, volumetric data, high-level graph structures, and surface representations such as meshes. This versatility makes point clouds a fundamental abstraction for 3D processing and analysis across various applications. In this work, we focus exclusively on geometric (spatial) noise affecting the positional accuracy of 3D points; noise solely in non-geometric attributes such as color, intensity, or reflectance, while relevant, lies outside the scope of this study.

Adding to these challenges, processing 3D data with complex geometries often incurs high computational costs, limiting real-time applicability. Addressing these issues requires robust adaptive frameworks capable of systematically classifying and mitigating diverse noise sources while preserving critical structural information. This work proposes a comprehensive taxonomy and survey aimed at unifying current knowledge and guiding future research in noise-adaptive 3D point cloud denoising methods.

In addition, this work aims to classify noise in 3D data along three main axes—statistical, environmental, and structural—and to highlight effective strategies for its mitigation. Our taxonomy

organizes fragmented knowledge into a unified framework, enabling the design of robust, adaptive systems for noise handling in diverse 3D sensing applications.

2. Taxonomy of Noise in 3D Data

2.1. Classification Criteria

With the growing adoption of three-dimensional data to tackle challenges once addressed solely through images, the landscape of 3D sensing technologies has expanded at an unprecedented pace. Each sensor, with its own physical principles and limitations, imprints a unique signature on the data it acquires. Combined with the randomness of environmental conditions and other unpredictable agents, this results in an inherent anti-determinism in 3D acquisition. Yet, within this diversity lies recognizability: noise often carries statistical fingerprints and discernible patterns that, once understood, reveal its origin. Consequently, while some denoising strategies are crafted within the framework of a particular sensor or scenario, others embrace generality, striving to enhance data quality across multiple acquisition technologies.

To establish a systematic and reproducible classification framework, we define three fundamental criteria that enable the categorization of any noise type encountered in 3D point cloud processing. These criteria ensure that future noise models can be consistently classified within our taxonomy.

1. **Origin-based Criterion:** Distinguishes noise sources according to their fundamental cause:
   - *Intrinsic*: Noise inherent to the mathematical/statistical nature of the measurement process.
   - *Extrinsic*: Noise caused by external environmental factors during acquisition.
   - *Systemic*: Noise introduced by the acquisition system, processing pipeline, or reconstruction algorithms.
2. **Modeling Criterion:** Evaluates the mathematical tractability and predictability of noise behavior.
   - *Parametric*: Noise that can be described by well-defined probability distributions with estimable parameters.
   - *Deterministic*: Noise with predictable patterns that can be modeled algorithmically.
   - *Non-parametric*: Noise that requires data-driven or empirical modeling approaches.
3. **Spatio-Temporal Dependency Criterion:** Characterizes the correlation structure of noise:
   - *Independent*: Noise affecting individual points without spatial or temporal correlation.
   - *Structured*: Noise that exhibits spatial patterns, temporal evolution, or sensor-specific regularities.
   - *Global*: Noise affecting entire acquisitions or large regions uniformly.

These criteria provide a decision framework where any noise type can be systematically assigned to one of three main categories based on its predominant characteristics. The intersection of these criteria guides the placement within our three-axis taxonomy.

2.2. Statistical Noise Models

This category refers to noise types that arise from the fundamental statistical nature of the measurement process and can be characterized by probability distributions. However, it is important to clarify that these noise models themselves are mathematical constructs designed to approximate the behavior of such noise, rather than phenomena that originate directly from the statistical nature of the process. These noise models generally satisfy the intrinsic origin and parametric modeling criteria and typically exhibit independent spatial behavior. For example, certain noise phenomena, such as those caused by sensor malfunctions, can be effectively approximated by impulsive noise models, even though such models do not represent physically existing noise types.

First, let a 3D point cloud without noise be defined as:

$$\mathcal{P} = \{\mathbf{p}_i \in \mathbb{R}^3 \mid i = 1, 2, \ldots, N\}$$

2.2.1. Gaussian Noise

Gaussian noise represents the most common statistical noise model in 3D acquisition systems, arising from thermal fluctuations in sensors, quantization errors, and multiple independent error sources following the Central Limit Theorem.

Mathematical Model: Noise follows a multivariate normal distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where typically $\boldsymbol{\mu} = \mathbf{0}$ for zero-mean noise. For point clouds, this results in:

$$\mathbf{p}_i' = \mathbf{p}_i + \boldsymbol{\varepsilon}_i, \quad \text{with} \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}), \tag{1}$$

where $\boldsymbol{\varepsilon}_i$ is additive, spatially uncorrelated and white (spectrally flat) when $\boldsymbol{\Sigma}$ is diagonal.

Gaussian noise is said to be **isotropic** (non-directional) when the variance is equal along all axes, i.e., $\sigma_x = \sigma_y = \sigma_z$, which leads to a covariance matrix of the form:

$$\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_{3\times 3}$$

This implies rotational invariance; i.e. the noise is distributed equally in all directions, producing a spherical distribution of perturbations around each point. As a type of noise that arises from statistical measurement processes and is characterized by a normal distribution with isotropic covariance, it can be categorized as *Intrinsic*, *Parametric*, and *Independent*.

Otherwise, if the variances differ per axis ($\sigma_x \neq \sigma_y \neq \sigma_z$), but there are still no inter-axis correlations, the noise is said to be **anisotropic** (directional), and the covariance matrix becomes diagonal:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_x^2 & 0 & 0 \\ 0 & \sigma_y^2 & 0 \\ 0 & 0 & \sigma_z^2 \end{bmatrix}$$

This models ellipsoidal noise distributions aligned with the coordinate axes, but still without directional coupling.

More generally, the covariance matrix of a 3D Gaussian noise distribution can be expressed as:

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_x^2 & \rho_{xy}\sigma_x\sigma_y & \rho_{xz}\sigma_x\sigma_z \\ \rho_{xy}\sigma_x\sigma_y & \sigma_y^2 & \rho_{yz}\sigma_y\sigma_z \\ \rho_{xz}\sigma_x\sigma_z & \rho_{yz}\sigma_y\sigma_z & \sigma_z^2 \end{bmatrix},$$

where $\sigma_x^2, \sigma_y^2, \sigma_z^2 > 0$ represent the variances along each axis, and ρ_{ij} are correlation coefficients that satisfy $-1 \leq \rho_{ij} \leq 1$. This matrix is symmetric ($\Sigma_{ij} = \Sigma_{ji}$) and positive definite, ensuring a valid covariance structure.

The presence of non-zero off-diagonal terms encodes correlations between the axes, which means that the noise distribution may be oriented along directions that do not align with the Cartesian axes. Even if the variances are equal (i.e., the diagonal entries are the same), these correlations break the rotational symmetry of the distribution, making the noise **non-isotropic**.

Consequently, a specific example of a full covariance matrix used in [2] is:

$$\boldsymbol{\Sigma} = s^2 \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{4} \\ -\frac{1}{2} & 1 & -\frac{1}{4} \\ -\frac{1}{4} & -\frac{1}{4} & 1 \end{bmatrix}$$

Although the variances along each axis are equal (diagonal terms), the non-zero off-diagonal elements introduce cross-axis dependencies. Consequently, the resulting noise distribution is directionally biased and no longer spherically symmetric, hence *non-isotropic* in nature. In this case, the noise becomes *Structured*.

This general formulation offers a more accurate real-world noise approximation, as practical systems often exhibit correlated noise across dimensions due to shared mechanical or electronic components.

Moreover, empirical evidence supports the Gaussian approximation of Kinect-like depth sensor noise. [3] developed a spatial uncertainty model for Kinect sensors, representing measurement error with a full covariance matrix in Cartesian space—effectively modeling noise as multivariate Gaussian. Additionally, [4] proposed a noise measurement model for depth cameras (including structured-light and ToF sensors), where noise in the floor-plane domain is explicitly treated as a 2D Gaussian distribution.

Although most academic articles place particular emphasis on testing models and comparing their performance against synthetic Gaussian noise with zero mean and isotropic standard deviation, this, like other simple parametric distributions, does not faithfully represent complex real-world noise [5,6].

2.2.2. Poisson Noise

Poisson noise typically applies to discrete measurements and is dependent on the signal (amplitude), unlike Gaussian noise, which is additive and signal-independent. It is more suitable for modeling noise in the signal attributes of a point cloud, such as color or intensity, rather than the geometric coordinates that are continuous rather than discrete counts. Emerging from intrinsic signal-dependent processes, this parametric model describes discrete attribute variations through independent Poisson distributions.

Mathematical Model: The noise is based on the Poisson distribution with parameter λ_i, where the probability of observing k events is given by the following probability mass function (PMF):

$$\text{Poisson}(\cdot) \equiv P(k_i = k) = \frac{\lambda_i^k e^{-\lambda_i}}{k!}, \quad k = 0, 1, 2, \ldots$$

The expected value and variance of the Poisson distribution are:

$$\mathbb{E}[k_i] = \text{Var}(k_i) = \lambda_i$$

For a true signal attribute value s_i (e.g., intensity or color) of a point, the noisy observation s_i' can be modeled as:

$$s_i' \sim \text{Poisson}(s_i),$$

where the noise is signal-dependent and non-additive, with variance equal to the mean. Unlike Gaussian noise, it is not spatially uncorrelated in amplitude due to its dependence on local signal intensity.

Poisson noise is ubiquitous in imaging systems [7–10]. The work in [11] shows that the BM3D (3D Block Matching) filter—originally designed for Gaussian noise removal in 2D images—can be effectively adapted to Poisson noise using variance-stabilizing transformations (VSTs). Although their method operates on pixels, its core idea of grouping similar 3D patches (2D + time/depth) and applying collaborative filtering could inspire analogous approaches for 3D point clouds. In this context, "patches" might correspond to local geometric neighborhoods around each point, with similarity defined via local descriptors (e.g. normals, curvature) or attributes like intensity or color. Indeed, previous methods such as [12] have explored similarities between non-local surface descriptors for denoising 3D unstructured data.

2.2.3. Laplacian Noise

Laplacian noise is a type of additive noise characterized by heavier tails than Gaussian noise, which makes it more robust to outliers. As defined by its intrinsic heavy-tailed properties, this parametric noise exhibits independent additive perturbations via Laplace distributions.

Mathematical Model: The noise vector $\boldsymbol{\varepsilon}_i$ added to a point $\mathbf{p}_i$ follows a multivariate Laplace distribution with zero mean:

$$\mathbf{p}_i' = \mathbf{p}_i + \boldsymbol{\varepsilon}_i, \quad \text{where} \quad \boldsymbol{\varepsilon}_i \sim \text{Laplace}(\mathbf{0}, b),$$

where the components are typically independent (spatially uncorrelated) but exhibit stronger deviations than Gaussian noise.

For the univariate case, the Laplace distribution with mean and scale parameter $b > 0$ has the following probability density function (PDF):

$$P(x) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right)$$

For multidimensional data, the noise components are often assumed to be independent and identically distributed Laplace variables on each axis.

Laplacian noise is useful when the noise model must account for frequent small errors plus occasional large jumps, unlike Gaussian noise, which heavily penalizes large deviations.

Experimental tests with Laplacian noise are conducted in [13], where it is highlighted that, unlike Gaussian noise, Laplacian noise is sparse but contains larger outliers. Therefore, it is essential to design robust methods that can handle these outliers effectively, allowing for more efficient removal of Laplacian noise without excessively smoothing out the fine details of the point cloud.

Acquired 3D data typically contain small positional errors that can be approximately modeled as zero-mean additive noise—either Gaussian (frequent small errors) or Laplacian (sporadic larger errors). As noted in [13], while complex sensor-specific noise models exist, simpler approximations such as those presented there balance accuracy with computational feasibility. Those extend prior Gaussian-only approaches and establish a more rigorous framework for handling both types of noise.

2.2.4. Impulsive Noise/Outliers

Impulsive noise is non-additive and spatially sparse. Also known as salt-and-pepper noise or outliers, it is characterized by occasional large deviations that do not follow a smooth distribution like Gaussian noise. Resulting from intrinsic sensor failures, this parametric noise independently replaces points with outliers via Bernoulli-uniform sampling or analogous methods.

Mathematical Model: The noisy point $\mathbf{p}_i'$ can be modeled as:

$$\mathbf{p}_i' = \begin{cases} \mathbf{p}_i, & \text{with probability } 1-\alpha \\ \mathbf{o}_i, & \text{with probability } \alpha, \end{cases}$$

where $\mathbf{p}_i$ is the true point coordinate, $\mathbf{o}_i$ is an outlier point (directionally unbiased or uniformly distributed unless constrained to specific axes), typically drawn from a uniform or other broad distribution (i.e. a distribution with wide spatial support) over the 3D space, and $\alpha \in [0, 1]$ is the probability of a point being corrupted by the noise.

This model assumes that most points are unchanged, but a fraction α are replaced by outliers randomly distributed in space.

In real-world captures, impulsive noise frequently appears in spatial bursts (e.g., contiguous outlier clusters from transient occlusions or sensor glitches). This suggests extending the Bernoulli model to Markovian frameworks, where outlier probability depends on neighboring point states—akin to Markov-Middleton models in signal processing. Moreover, bursty outliers violate the *i.i.d.* assumption, necessitating denoising methods that account for local spatial correlations, such as Markov random fields (MRFs) or density-aware clustering [14].

2.2.5. Speckle Noise

Speckle noise is one of the most important forms of multiplicative and spatially correlated noise, so it follows a multiplicative model:

$$w(\mathbf{x}) = s(\mathbf{x}) \cdot n(\mathbf{x}),$$

where $s(\mathbf{x})$ denotes the clean signal (e.g., geometry or attributes of points), $n(\mathbf{x})$ is the noise field, and $w(\mathbf{x})$ are the noisy data observed.

As this type of noise is modeled using algorithmic Perlin flow fields, generating intrinsic multiplicative and spatially coherent distortions, it can be categorized as deterministic and structured.

Mathematical Model: To synthesize speckle noise in point clouds, a common strategy is to use *Perlin noise*, a spatially coherent, gradient-based pseudo-random function originally developed for texture synthesis. Speckle is simulated by applying a Perlin noise field multiplicatively over the surface:

$$\mathrm{PN}(x, y) = \sum_{m=1}^{N} A_m \cdot N_m(f_m x, f_m y),$$

where N_m denotes the m-th octave of the noise function, with frequency $f_m = f_0 \cdot 2^{m-1}$ and amplitude $A_m = A_0 \cdot p^{m-1}$. Parameters f_0, A_0, and persistence $p \in (0, 1)$ control the level of detail and spatial variation [15]. Essentially, it introduces anisotropic structured perturbations aligned with the surface.

This model yields spatially structured fluctuations that mimic real-world speckle effects and provides a controllable way to introduce non-additive, anisotropic distortions into 3D data.

2.2.6. Uniform Noise

Uniform noise is an additive and bounded noise model commonly used in synthetic experiments to test denoising methods under intrinsic limited but random perturbations. It is characterized by having an equal probability for all values within a fixed interval. This parametric model applies independent uniform perturbations within fixed intervals, representing random uncertainties.

Mathematical Model:

$$\varepsilon_i \sim \mathcal{U}(a, b), \quad p(\varepsilon) = \begin{cases} \frac{1}{b-a}, & a \leq \varepsilon \leq b, \\ 0, & \text{otherwise}, \end{cases}$$

making it non-directional if symmetric ($a = -b$) or directional if biased (e.g., $a = 0$).

When applied to 3D point clouds, each point variable is independently perturbed, and it serves as a useful baseline for controlled noise studies due to its well-defined bounds and simplicity.

The method in [16] shows strong performance against uniform noise (generated by a 3D ball distribution) through its gradient field learning approach. Quantitative evaluation shows good Chamfer Distance and Point-to-Mesh metrics. The key advantages are that the gradient field estimation naturally pulls uniformly distributed points toward the underlying surface without explicit noise modeling, that the continuity-preserving resampling algorithm maintains geometric consistency during denoising and that The method generalizes well to unseen noise types despite being trained only on Gaussian noise. The results confirm the robustness of their physics-inspired approach to uniform noise removal while preserving surface details.

Another physics-inspired approach to uniform noise removal is similar to the approach of [17], which demonstrates effective uniform noise suppression in 2D vector fields through physical constraints (solenoidal projection removing divergent noise components), spectral filtering (preserving large-scale flow structures while attenuating high-frequency noise) and vorticity rectification (enforcing physical consistency). These principles can be directly translated to 3D point cloud denoising by treating point displacements as vector fields and the underlying surface as a physical constraint. The projection operation becomes surface reconstruction (e.g. via MLS or RANSAC), spectral filtering translates to neighborhood-based smoothing that preserves geometric features while reducing random displacements, and the vorticity constraint corresponds to normal/curvature regularization that pulls noisy points toward geometrically consistent positions. This physical-to-geometric analogy provides

a robust framework for correcting 3D point displacements while maintaining surface continuity, mirroring how fluid dynamics maintains flow continuity in 2D denoising.

The method proposed in [18], based on L^∞ norm fitting, is particularly well-suited for inverse problems with this type of noise, as it explicitly accounts for the noise's bounded support. as the functional constraint focuses on the largest values that the signal takes, ignoring the small fluctuations, and ensuring that the reconstruction tolerates the worst-case perturbation without over-smoothing edges or textures. This framework extends seamlessly to 3D point clouds, where uniform noise arises naturally in spatial coordinates. Here, each point is perturbed independently within a hypercube $[-a, a]^3$, and the L^∞ residual minimization enforces geometric consistency by bounding the maximum deviation across all dimensions. Unlike L^2 methods, which are sensitive to outlier perturbations, the L^∞ approach guaranties uniform data fidelity.

2.2.7. Gamma Noise

Gamma noise is suitable for modeling non-negative, skewed deviations and is relevant in scenarios involving multiplicative effects and photon statistics, typical in laser-based imaging systems, offering a more realistic representation than symmetric additive models. This type of noise originates from intrinsic photon statistics, and parametrically models skewed positive deviations via Gamma distributions.

Mathematical Model:

$$\varepsilon_i \sim \text{Gamma}(k, \theta), \quad p(\varepsilon) = \begin{cases} \frac{\varepsilon^{k-1} e^{-\varepsilon/\theta}}{\theta^k \Gamma(k)}, & \varepsilon \geq 0, \\ 0, & \text{otherwise}, \end{cases}$$

where k is the shape parameter, θ is the scale parameter and $\Gamma(\cdot)$ is the Gamma function. This distribution is also known as Erlang noise in the laser imaging context.

2.2.8. Rayleigh Noise

Rayleigh noise arises in magnitude measurements involving coherent signals, such as radar and range imaging, where the underlying complex signal components may be modeled as Gaussian distributions. It is produced by intrinsic magnitude measurements in coherent systems; this parametric noise follows independent Rayleigh distributions.

Mathematical Model:

$$\varepsilon_i \sim \text{Rayleigh}(\sigma), \quad p(\varepsilon) = \begin{cases} \frac{\varepsilon}{\sigma^2} e^{-\varepsilon^2/(2\sigma^2)}, & \varepsilon \geq 0, \\ 0, & \text{otherwise}, \end{cases}$$

where σ is the scale parameter of the noise distribution. This model is often applied to noise in range imaging and radar magnitude data. Rayleigh noise is skewed and strictly positive, capturing the distribution of envelope fluctuations.

2.2.9. Unidirectional Noise

This noise model is additive and axis-aligned, perturbing only one coordinate. It tests the ability of denoising algorithms to handle noise with directional bias and a reduced complexity structure. As it independently adds axis-aligned Gaussian perturbations, this parametric noise simulates intrinsic anisotropic effects.

Mathematical Model: Unidirectional noise restricts perturbations to only one coordinate axis, modeling anisotropic noise that affects measurements primarily along a single direction. This can represent physical scenarios where sensor inaccuracies or environmental factors induce noise predominantly along one axis.

In practice, this is often implemented by adding Gaussian noise only to a single component of each point. For example, the x-component is perturbed as:

$$p_i' = p_i + (\varepsilon_i, 0, 0), \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where σ is set as a relative percentage of the bounding region (e.g., the radius of the bounding sphere or the side length of the bounding box) to control the noise intensity. In [2], noise levels of 1%, 2%, and 3% are used.

2.3. Scene-Induced Interferences

This category includes noise that arises from environmental factors and scene characteristics that interfere with the acquisition process. These noise types satisfy the *extrinsic origin* criterion and typically exhibit *deterministic* or *structured* spatial-temporal dependencies.

2.3.1. Specular/Reflective Surfaces

Reflective surfaces, such as polished metal, ceramics, or translucent materials, are a significant challenge in 3D scanning due to the way they interact with light, e.g., with laser or natural light. Instead of simply bouncing coherently back to the sensor, these surfaces can scatter or reflect in unexpected and random ways, altering the resulting 3D data. It is categorized as extrinsic (environmental origin), non-parametric (empirical mitigation), and structured (spatial coherence created by complex light-scattering artifacts).

In [19] a common issue is described in this respect, which is the presence of long-range outlier noise, i.e. points that appear noticeably far from the true surface. Such outlier-like noise usually results from single reflections, when the laser hits the object at sharp angles, or multiple reflections, when the scanned surface is characteristically shiny. The errors derived from these phenomena can be grouped into two categories: (1) Isolated noise points, which are relatively easy to remove using filters that detect distant anomalies, and (2) Attached noise clusters, which lie close to the surface and are much harder to remove without altering the true geometry of the object.

But even when the reflectivity involved is not extreme, it is typical that subtle variations in how the light reflects on the surface still cause short-range, high-frequency noise. This distortion usually blurs fine details, but without affecting the overall shape, as also noted in [19].

Another important phenomenon is edge flare—a known effect that occurs near sharp edges or corners (i.e. geometric discontinuities). In these zones, the sensor tends to lose accuracy because of the sharp changes in geometry, scattering light unpredictably, making it impossible in practice to acquire a measurement.

In addition, Multi-view stereo (MVS) systems face additional challenges. This is because, according to [20], reflections can change depending on the viewpoint, creating specular highlights that confuse stereo matching procedures and algorithms. This results in structured clusters of false points that may be spatially coherent but do not correspond to the actual surface, worsening the already low reliability of these reconstruction methods for high-precision applications.

The analysis of highly reflective surfaces such as solar panels, polished metals, and ceramics presents unique imaging challenges, as investigated in [21]. These materials exhibit three characteristic difficulties: intense specular reflections that mask genuine defects and generate false artifacts, extreme sensitivity to environmental conditions where minor lighting or viewpoint changes dramatically alter surface appearance, and complex microstructural properties that introduce high-frequency noise patterns. The combination of these factors renders conventional defect detection approaches ineffective, as they struggle to distinguish between actual surface imperfections and reflection-induced artifacts.

[21] addresses these challenges through an improved Faster R-CNN architecture featuring three principal innovations. First, the framework employs adaptive anchor boxes optimized for micro-defect detection via K-Means clustering. Second, it implements a hierarchical feature fusion approach that combines multi-scale visual information to differentiate true defects from reflection noise. Third, the system incorporates illumination-invariant training protocols to maintain stability in varying lighting conditions. While originally developed for defect identification, this methodology's robust handling of

reflective surface artifacts suggests promising applications in 3D point cloud denoising, particularly for scenarios involving specular materials where traditional noise removal techniques often fail.

Finally, translucent materials introduce yet another layer of difficulty. In this sense, light does not just reflect off them, but it penetrates and scatters within. [19] highlights that this internal scattering, combined with external reflections, creates both long-range outliers and dense patches of short-range noise, seriously compromising the quality of CAD or metrology scan scenarios.

2.3.2. Dynamic Obstructions

Dynamic obstructions refer to transient, moving objects—such as pedestrians, vehicles [22], or debris [23]—that interrupt 3D scanning and appear as spurious clusters of outliers. Unlike static noise, these obstructions manifest briefly and unpredictably, often creating temporary point clusters or ghost points that drift in and out of the scanned data. These artifacts require multi-frame, time-aware filtering strategies, since single-frame denoising is usually insufficient. As noted in [24], these artifacts manifest themselves as structured but transient anomalies that confuse downstream tasks such as semantic segmentation and object tracking, which makes it vital to suppress them. In this sense, it classified under extrinsic sources with non-parametric modeling and structured dependencies, as transient objects create spatiotemporal outlier clusters.

Although they form clusters similar to surface outliers, dynamic obstructions differ because they disappear suddenly, introducing temporal discontinuities in the noise pattern. This erratic behavior sets them apart from other ones like persistent geometric noise or seasonal environmental artifacts. Moreover, each point cloud frame is usually irregularly sampled, and different frames may exhibit varying sampling patterns with possibly different numbers of points, which means there is no explicit temporal correspondence between points over time, which complicates the task [25].

Dynamic obstructions are particularly problematic because they do not maintain consistency over time. Their presence in one frame and absence in the next can create discontinuities that are difficult to distinguish from genuine scene features or legitimate structures. This requires the development of specialized techniques to effectively identify and mitigate their impact.

Additionally, **motion blur** artifacts—caused by relative movement between the sensor and objects during acquisition—can manifest as elongated point clusters or smeared surfaces. These effects share similarities with dynamic obstructions, but arise from temporal integration rather than physical occlusions, requiring motion-aware reconstruction or deblurring techniques.

Furthermore, [22] highlight an additional issue with Terrestrial Laser Scanning (TLS) point clouds. In uncontrolled scenes, the scanned data often suffer from irregular data gaps due to moving objects like pedestrians or cars, leading to a loss of integrity and quality, and since dynamic occlusions are transient, they create gaps that must be identified and filled in order to achieve accurate reconstructions. This makes it essential to develop effective techniques to handle such cases.

2.3.3. Light Interference

Light scattering can also cause the appearance of long-range noise in scanned objects [19]. A well-known effect is edge flare, which refers to aggravated errors at sharp edges or corners (geometric discontinuities) caused by light scattering. This phenomenon leads to a loss of measurement accuracy in these regions due to the sensor's reduced ability to capture precise data, all due to the lighting or reflective nature of the surface or artifact in the scene [26].

So, exhibiting extrinsic, non-parametric, and structured traits, scattered light usually induces edge flare artifacts at geometric discontinuities.

2.3.4. Temperature Variations

Temperature variations are an extrinsic and deterministic noise source that globally affects 3D point cloud data, particularly in sensor-based acquisition systems. These thermal shifts cause systemic calibration drifts in LiDAR or structured-light sensors, leading to inaccuracies in depth measurements and introducing distortions in the captured 3D surface geometry.

Unlike random noise, temperature-induced errors follow predictable patterns but require specialized denoising approaches to correct without compromising fine structural details. As demonstrated in [27], temperature fluctuations produce distinct noise patterns in 3D scans, where the authors use statistical thresholds to automatically detect and filter these thermal artifacts, enabling accurate surface reconstruction while maintaining critical micro-texture details.

The method in [27] tackles temperature-related 3D noise using a two-step approach. First, it applies a modified Gaussian filter that ignores extreme values (common in temperature errors) by setting the kernel's center weight to zero. Second, it detects outliers, points with values three times higher than average, and replaces them with neighborhood averages. It also uses coefficient of variation (CV) analysis to spot smaller local errors without affecting real surface details. Tests prove this method outperforms traditional filters on real surfaces, it preserves original texture better while effectively removing temperature errors. It handles both gradual temperature changes (with the Gaussian filter) and sudden sensor errors (with CV detection).

2.3.5. Vibrations or Mechanical Instability

Demonstrating extrinsic, deterministic, and structured behavior, mechanical resonances produce spatially correlated distortions in the point cloud. This noise source is prevalent in mobile mapping systems, such as UAVs or handheld scanners, where the platform's engine or movement introduces high-frequency oscillations to the sensor's pose during acquisition.

Unlike random noise, these vibrations do not affect points independently. Instead, they manifest as wave-like patterns or "ripples" on otherwise planar surfaces, strictly following the scanning trajectory. Since the error depends on the platform's oscillation frequency and the specific timestamp of the measurement, the resulting distortion is highly structured and predictable, often requiring trajectory smoothing or inertial data fusion for correction rather than simple local averaging.

2.3.6. Adverse Weather

Qualified as extrinsic, non-parametric, and structured noise, adverse weather conditions, such as atmospheric particles, rain, fog, and snow, create localized interference patterns and introduce noise into scanned point clouds, significantly degrading their quality and reliability for downstream perception tasks in autonomous driving and robotics. Environmental particles scatter, absorb, or anomalously reflect laser pulses (for example), generating false returns and occluding true surface geometries. Unlike intrinsic sensor noise (e.g., Gaussian noise), adverse weather effects exhibit spatial and temporal correlations, making them challenging to filter using traditional statistical or geometric heuristics.

As noted in [24], the noise characteristics vary significantly between different weather conditions. For example, fog produces dense, low-frequency scattering that obscures distant objects and attenuates intensities; rain introduces localized, high-frequency noise with sparse but persistent outliers; and snow generates spatially clustered interference caused by large, reflective particles, often mimicking valid surfaces. In [28], intensity-dependent spatial correlations are leveraged to distinguish fog particles from valid surfaces, while TripleMixer employs a hybrid geometric–spectral approach to decouple weather-induced artifacts from true geometries in both the spatial and frequency domains.

The development of large-scale datasets (e.g. Weather-KITTI, Weather-NuScenes, etc. [24] with point-level weather annotations has further enabled data-driven solutions, overcoming the limitations of synthetic simulations and small-scale real-world collections.

2.4. Acquisition/Reconstruction Imperfections

This category encompasses noise introduced by the acquisition system itself, reconstruction algorithms, and processing pipelines. These noise types satisfy the *systemic origin* criterion and often exhibit *structured* or *global* spatial dependencies.

2.4.1. Structured/Sensor-specific Noise

Structured noise commonly originates from real LiDAR-type sensor acquisitions, which is why it is also referred to as sensor-specific noise. It is characterized by depth-dependent, angular, or sparse perturbations. Empirical studies such as [29] have demonstrated that the measurement noise in LiDAR sensors increases with distance due to a lower signal return strength, resulting in range-dependent noise patterns. It also suffers from angular noise caused by the discrete scanning mechanism. The study by [30] further shows that different commercial sensors exhibit distinct error patterns depending on the model, distance, incendence angle or the type of material being scanned. These errors are not random, but systematic and depend on the sensor, the surface, and lighting conditions.

Beyond intrinsic sensor limitations, environmental and operational factors can exacerbate structured noise. Low illumination, high ambient or sensor temperatures, dust or airborne particles, and transmission interference may degrade the measurements. Such conditions can introduce impulsive noise, generate speckle from wave backscattering, or produce *Poisson-like* noise from the stochastic nature of photon arrival combined with nonlinear sensor responses. These effects yield deterministic or semi-deterministic patterns with spatial correlation, characteristic of sensor-specific noise.

In [31], they attempt to generalize Gaussian noise and address structured noise at high levels. Specifically, they train their model with simulated noise on a LiDAR dataset by scanning ShapeNet objects with a Velodyne HDL-64E scanner using Blensor software [32]. They consider two sources of noise: a laser distance bias modeled as a Gaussian distribution and Gaussian noise added per laser beam (both distributions are set with zero mean and a standard deviation proportional to the region of interest). Similarly, in [16], Blensor is also used to simulate LiDAR noise. They first construct a mesh from the original point cloud using MeshLab, normalize it into a unit sphere, and scan it from six directions with a Velodyne HDL-64E2 model and different noise levels.

Similarly, stripe noise in 2D images is widely recognized as structured noise, as noted by [33], while [34] extend this understanding to 3D imaging, categorizing it as sensor-specific noise arising from hardware imperfections or interference. This noise manifests itself as depth/angle-dependent stripe patterns with varying widths and intensities, systematically degrading image quality due to deterministic sensor limitations. To address this, [34] propose a dual-resolution 3D U-Net model (USRL) that directly learns and removes multiscale 3D stripes through hierarchical feature extraction, coupled with nonlinear dynamic range normalization to adapt to diverse stripe intensities. Their approach outperforms conventional methods (e.g. DnCNN [35]) by preserving structural details while eliminating artifacts, as demonstrated in LSFM neural data, and offers extensibility to other modalities like infrared or medical imaging despite current limitations in processing wide stripes or low-light conditions.

Two additional sensor-specific noise types are **shot noise** and **quantization noise**. Shot noise, fundamental to photon-counting sensors like LiDAR, follows Poisson statistics and becomes dominant under low-light conditions, creating signal-dependent variance that scales with measurement intensity. Quantization noise represents systematic errors introduced by analog-to-digital conversion, particularly impactful in time-of-flight (ToF) sensors where depth discretization creates stair-step artifacts at quantization boundaries. Both noise types exemplify how sensor physics imposes structured error patterns that require specialized modeling for effective mitigation.

2.4.2. Data Gaps

Resulting from systematized occlusions or acquisition failures, these deterministic artifacts produce missing-structured regions that often follow the scanning geometry or motion pattern. Such gaps can severely hinder subsequent processing steps by breaking surface continuity and removing critical geometric cues.

The method proposed in [23] addresses pronounced data gaps (structured missing regions) or open meshes in the scans. Using discrete isotropic and anisotropic operators with geometry-aware edge weights, it extrapolates missing geometry while preserving sharp features. Anisotropic weights smooth

normals adaptively, and a bi-tensor voting scheme reliably detects edges and corners despite gaps. A RANSAC-based multi-normal estimation discards outliers and infers plausible surface orientations, improving robustness to incomplete data. Validated on synthetic Gaussian noise, impulsive noise, and unorganized point sets, the method processes raw and irregularly sampled clouds directly with KNN-based operators, avoiding reliance on global topology. Tests on Kinect data show tolerance to real-world imperfections, such as holes, stitching artifacts, and partial views, while a unit-normal-constrained optimization ensures consistency in incomplete regions.

In addition, TP-NoDe [36] handles data gaps using its noise-denoising upsampling approach. The method injects controlled noise into sparse regions and applies denoising to reconstruct the underlying surface geometry. This process naturally fills missing areas while preserving sharp features and surface continuity. The progressive refinement strategy enables effective reconstruction of incomplete scans without requiring explicit gap-filling algorithms or complex preprocessing steps.

2.4.3. Scan Stitching Errors

Scan stitching errors manifest as misalignments and geometric inconsistencies when fusing data from multiple depth sensors, such as in multi-camera Poisson reconstruction setups. These errors arise from calibration inaccuracies, temporal synchronization issues, and sensor-specific noise patterns.

These errors often create "balloon-like" artifacts in empty regions where depth measurements conflict, or strongly directional and artificial patterns and clusters of outliers, as shown in [37].

Although traditional approaches often address these problems through post-hoc point cloud denoising, recent methods—such as [38]—focus on improving calibration and 3D data recovery during multi-view reconstruction itself. By refining alignment parameters through Kalman filtering and leveraging auxiliary data (e.g., reflectance images or geometric constraints), these techniques reduce stitching errors at the source rather than correcting them after the fact, not only minimizing artifacts but also preserving structural integrity, particularly in large-scale scans where error accumulation is a major concern.

The work in [39] introduces an innovative stitching methodology for 3D point clouds using PDE-based analytical solutions. Their approach fundamentally addresses the stitching problem by enforcing geometric continuity through explicit PDE solutions that mathematically guaranty smooth transitions between stitched patches. Unlike conventional stitching techniques that rely on iterative alignment, this method formulates the stitching process as a boundary-value problem where the PDE solutions automatically maintain both positional and tangential continuity at stitch boundaries, although the implementation focuses on basic stitching scenarios (single-patch reconstructions).

2.4.4. Discrete Noise

Discrete noise refers to perturbations in which point displacements occur only at a finite set of fixed offsets rather than continuously across a range. This type of noise can model sensor errors that produce quantized, atypical, or degenerate measurements, often due to digitization or mechanical constraints. Modeling systemic digitization errors, this parametric type og noise applies independent quantized displacements.

Formally, the noise distribution is defined as a discrete probability mass function on a small set of displacements $\{\mathbf{d}_k\}$:

$$p(\mathbf{x}; s) = \begin{cases} p_k, & \mathbf{x} = \mathbf{d}_k(s), \\ 0, & \text{otherwise}, \end{cases}$$

where s is a scale parameter that controls the magnitude of the displacements and the weights p_k sum (ideally) to 1. Any displacement not explicitly listed among the $\mathbf{d}_k(s)$ has zero probability of occurring.

For example, in [2], the simplified discrete noise model perturbs points with probability 0.1 at offsets $\pm s$ along each coordinate axis, and with probability 0.4 leaves the points unchanged:

$$p(\mathbf{x}; s) = \begin{cases} 0.1, & \mathbf{x} = (\pm s, 0, 0) \text{ or } (0, \pm s, 0) \text{ or } (0, 0, \pm s), \\ 0.4, & \mathbf{x} = (0, 0, 0), \\ 0, & \text{otherwise.} \end{cases}$$

This noise model captures non-continuous, quantized perturbations that are realistic in certain acquisition scenarios.

3. Noise Mitigation Techniques

This section provides a comprehensive review of published denoising and filtering techniques organized by their applicability to the noise categories established in our taxonomy. We analyze computational complexity, parameter sensitivity, and cross-noise category effectiveness for each approach.

3.1. Gradient-Guided Point Set Resampling

[16] propose a probabilistic denoising framework that reformulates point cloud restoration as gradient ascent on the log-density of a noise-convolved distribution $p * h$. Their method demonstrates effective performance across multiple noise types, including isotropic Gaussian noise, Laplace noise, discrete noise, anisotropic Gaussian noise, uni-directional Gaussian noise, uniform noise, simulated LiDAR noise, and real-world noise. It particularly excels against structured parametric noise (e.g., anisotropic/discrete) and scene-induced interferences (e.g., LiDAR artifacts).

Statistical Noise Mitigation: For parametric noise models—such as isotropic Gaussian, Laplace, or uniform noise—the method's mode-seeking gradient ascent is theoretically grounded in the observation that the clean surface corresponds to the mode of $p * h$. By iteratively moving points along the gradient $\nabla_x \log[(p * h)(x)]$, it avoids shrinkage artifacts common in displacement-based methods (e.g., PCN [40]). This is especially effective for Laplace noise with heavy tails, where the continuity of the gradient field (enforced by cosine annealing) prevents over-smoothing while preserving outliers near the surface.

Structured and Scene-Induced Noise: The global gradient field estimation of the method proves critical for anisotropic Gaussian noise and discrete directional noise (both classified as structured). Unlike local operators (e.g., bilateral filtering), the global field aggregates features from all neighbors within the radius r, weighted by cosine annealing to ensure continuity. This allows the model to handle directionally varying corruption—such as anisotropic covariance or discrete spikes along axes—without assuming noise symmetry. For real-world LiDAR noise (a scene-induced interference), the adaptability of the gradient field to local geometry mitigates scanline artifacts and irregular shadows.

Systemic and Non-Parametric Noise: While the method is trained only on Gaussian noise, its regularization-flexible resampling generalizes to non-parametric noise (e.g., Paris-rue-Madame real-world data). By alternating gradient ascent with Graph Laplacian Regularization (GLR) or Reweighted GLR (RGLR), it adapts to unstructured corruption during inference. RGLR's dynamic edge weights are pivotal for acquisition imperfections, such as inhomogeneous density or sensor-specific artifacts, as they iteratively refine the underlying graph to preserve sharp features.

Computational Trade-offs: Iterative resampling (50+ steps) incurs higher costs for global noise (e.g., uniform spatial corruption) compared to feedforward networks. However, the authors argue that this is justified for structured noise where local methods fail. The continuity constraint reduces the Chamfer distance by 12% for anisotropic noise by preventing abrupt gradient changes near the surface boundaries.

Most of the time, this framework excels where noise exhibits: directional structure (anisotropic/discrete), due to global gradient estimation; parametric heavy tails (Laplace), through robustness to outliers; and scene/systemic patterns (LiDAR/reconstruction artifacts), through adaptive regularization. Its limitations comprise computational cost and initialization sensitivity.

3.2. Graph-Convolutional Point Denoising Network

Structured Systemic Noise: GPDNet [41] presents a method with distinct architectural advantages that make it particularly effective against different types of noise. Its performance is most notable when dealing with *structured systemic noise*, such as the spatially correlated artifacts found in LiDAR acquisitions. The network excels in this category due to its dynamic graph construction, which continuously updates neighborhood relationships in the feature space rather than relying on fixed 3D coordinates. This allows GPDNet to adapt to sensor-specific noise patterns, such as laser distance bias or per-ray Gaussian perturbations. Additionally, its non-local receptive fields, which expand progressively from 16 to 189 points, enable the capture of spatially dependent noise, making it robust against systemic imperfections in real-world 3D sensing systems.

Statistical Noise: For *statistical noise*, particularly Gaussian models, GPDNet employs a specialized loss function (MSE-SP) that combines point-to-point error with surface proximity regularization. This prevents common denoising artifacts, such as point clustering or filament structures, which often arise in learning-based approaches. The hierarchical feature learning—gradually expanding from 33 to 99 channels—further enhances its ability to filter out independent noise while preserving geometric fidelity. The adaptive neighborhood selection (defaulting to 16 nearest neighbors) ensures a balance between noise suppression and detail retention, particularly useful in high-noise scenarios.

Limitations: GPDNet is not equally effective against all types of noise. Its computational cost increases as a result of dynamic graph updates, which may limit real-time applications. Additionally, it shows reduced effectiveness against *environmental noise* (extrinsic category), such as occlusions or illumination artifacts, since these often require scene-specific context beyond geometric correction. The method also depends on careful parameter tuning, particularly in selecting the optimal neighborhood size, which can affect performance if not configured properly.

Overall, GPDNet aligns best with noise types that exhibit *systemic origin* (e.g., sensor-based distortions) and *structured dependencies* (e.g., LiDAR ray patterns). Its dynamic graph convolutions and feature-space processing make it a strong candidate for denoising tasks where traditional methods—reliant on surface fitting or fixed neighborhood assumptions—struggle. This positions GPDNet as a particularly effective solution for real-world 3D data corruption where noise follows predictable, acquisition-related patterns rather than purely random or scene-induced distortions.

3.3. Score-Based Point Cloud Denoising

Statistical Noise Performance: [2] demonstrates particular effectiveness against *statistical noise* due to its foundational probabilistic framework. By modeling the noisy point cloud as a convolution ($p * n$) between the clean surface distribution and noise distribution, the method naturally handles Gaussian noise through gradient ascent toward the distribution's mode. The local score estimation architecture preserves geometric features while filtering noise, avoiding common artifacts like shrinkage that plague displacement-based methods. The algorithm's training objective, which enforces score consistency across local neighborhoods rather than single points, enhances its robustness against various statistical noise models including non-Gaussian distributions.

Structured Noise Adaptation: While primarily designed for statistical noise, the method shows notable adaptability to *structured noise* through its ensemble scoring mechanism. The dynamic aggregation of neighborhood information (as nearest neighbors) allows handling of spatially correlated patterns like LiDAR artifacts, despite being trained solely on synthetic Gaussian noise. This generalizability stems from the local processing paradigm that learns fundamental geometric relationships rather than complete shapes, making it robust to acquisition-specific noise patterns.

Noise Limitations: The purely geometry-focused score estimation lacks semantic understanding to distinguish true surface points from obstructions or shadows. Performance also degrades on highly non-uniform noise distributions where the convolution model assumptions break down.

This method is optimally aligned with noise types that *exhibit intrinsic origin* (sensor measurement errors) and *independent / parametric* characteristics. Its theoretical grounding in distributional

modes provides unique advantages for statistical noise removal while maintaining adaptability to certain structured noise patterns. The local processing framework makes it particularly suitable for applications that require the preservation of fine geometric details in varied acquisition scenarios.

3.4. Density-based Denoising

The method proposed in [26] is particularly effective in handling *impulsive noise/outliers* and *structured / sensor-specific noise* in 3D point clouds. The method leverages kernel density estimation (KDE) with an optimized bandwidth selection via particle swarm optimization (PSO), making it robust against irregularly distributed outliers that deviate significantly from the underlying surface. The mean-shift clustering step efficiently isolates and removes these outliers by exploiting their low-density characteristics, which is ideal for impulsive noise where erroneous points are sparsely distributed, as in *light interferences*. Additionally, subsequent bilateral mesh filtering smooths high-frequency noise while preserving geometric features, addressing structured noise artifacts introduced by sensor imperfections.

Since the method relies on density-based outlier rejection, subtle noise patterns that do not significantly alter local point densities may remain unaddressed. However, the bilateral filtering step mitigates this limitation to some extent by smoothing high-frequency variations. The denoising pipeline excels in scenarios where noise manifests itself as sparse, high-magnitude deviations. Its adaptive bandwidth selection ensures robustness across varying noise intensities (density-aware outlier removal), while the bilateral filter preserves sharp features (feature-preserving smoothing). The framework is particularly suitable for raw point clouds from consumer-grade depth sensors.

3.5. 3D Local Shape Properties Feature Estimation

Statistical Noise and Outliers: PCN [40] demonstrates superior performance against *statistical noise*, particularly Gaussian distributions and impulsive outliers, which align with its training objectives. The method combines a classification network for outlier detection with an iterative denoising network that learns point displacements. This dual approach proves especially effective for independent noise satisfying the intrinsic origin criterion, as it does not require explicit noise modeling. The network's patch-based processing and multi-iteration refinement allow it to handle varying noise intensities without parameter adjustments, outperforming traditional methods like bilateral filtering and jet-fitting across different noise levels.

Structured Systemic Noise: For *structured systemic noise* including scanner-specific artifacts (e.g., Velodyne LIDAR's distance bias) and misaligned scans, PCN adapts through retraining on targeted datasets. Although not originally designed for these cases, its displacement-based denoising can correct anisotropic noise patterns and partially compensate for misalignments by learning to merge overlapping scans. However, its performance is highly dependent on the representativeness of training data, and it can preserve some structured artifacts when encountering unseen scanner configurations.

Environmental Noise Limitations: The approach shows reduced effectiveness against *scene-induce interference* categories such as occlusions or illumination artifacts [2], as these require scene context beyond local geometric correction. The method can cause point cloud shrinking—a common issue in similar methods. Additionally, its dependence on paired clean-noisy data for training restricts the model's ability to generalize to unseen or unmodeled noise distributions, limiting performance on structured or unknown noise types. The patch-based local approach may also struggle with global noise patterns and artifacts arising from misaligned scans. Finally, although the architecture avoids manual parameter tuning and is more efficient than graph-based methods, computational costs increase with point cloud density.

Noise robustness: PCN aligns best with noise types exhibiting *intrinsic origin* and *independent/parametric* characteristics, particularly Gaussian noise and sparse outliers. Its strength lies in handling acquisition imperfections that can be approximated through local displacement learning, making it particularly suitable for post-processing raw point clouds from various sensors when retrained on domain-specific data.

3.6. Mesh Total Generalized Variation

As noted in [42], while TGV [43] has proven effective for 2D tasks like speckle reduction and image restoration, its extension to 3D meshes presents unique challenges due to irregular connectivity. The authors address this through novel discrete operators that maintain TGV's feature-preserving qualities while handling mesh irregularity.

The Total Generalized Variation (TGV) based method proposed in [42] excels in handling structured and non-uniform noise in 3D meshes, particularly for surfaces combining smooth regions with sharp geometric features (e.g., CAD models). Unlike traditional approaches that prioritize homogeneous smoothing, the TGV regularizer operates at multiple orders, enabling adaptive edge preservation while effectively removing noise in homogeneous areas.

Optimal Noise Performance: The method demonstrates superior performance against high-variance Gaussian noise, where the spatial noise distribution may obscure fine features. The key lies in its dynamic weighting scheme, which iteratively adjusts weights based on normal differences between adjacent triangles. This attenuates noise in flat regions (low-frequency) without excessive smoothing of edges (high-frequency), even when the noise magnitude rivals feature size.

Key Advantages: The TGV method offers three key advantages for mesh denoising. First, it preserves sharp edges through the selective application of its first-order term near features while using the second-order term in smooth regions, avoiding the oversmoothing typical of the Laplacian-based (HO) [44] or TV [45] methods. Second, its discretized differential operators maintain robustness with irregular sampling and non-uniform mesh density, overcoming limitations of methods requiring regular structures such as NLLR [46]. Finally, the second-order formulation prevents artifacts common in L_0-based approaches [47], maintaining natural transitions in flat regions while eliminating stair casing effects.

Limitations: The method is computationally more expensive than alternatives such as BF or CNR, due to iterative linear system solving. Its efficacy diminishes when noise exceeds the size of geometric features, a common scenario with extreme outliers or impulsive noise. Nevertheless, unlike learning-based approaches, TGV's discrete operators generalize across CAD, non-CAD, and scanned surfaces, preserving sharp features while recovering smooth regions and showing superiority particularly in cases of heavy noise or complex geometric features.

TGV is ideal for environments where medium-to-high frequency additive noise must be removed without compromising inherent geometric structure, particularly in CAD models or scans with Gaussian artifacts. Its adaptability positions it as a versatile tool in reconstruction pipelines that prioritize geometric fidelity over processing speed.

3.7. Anisotropic Neighborhoods and Sharp Feature Detection

The proposed denoising method in [48] demonstrates particular effectiveness against additive noise due to its anisotropic neighborhood computation and bilateral normal filtering. The quadratic optimization formulation for neighborhood selection balances spatial proximity with normal variation, enabling selective smoothing that preserves sharp features while effectively reducing Gaussian-distributed perturbations. For impulsive noise and outliers, the algorithm's strength lies in its local neighborhood analysis and normal correction mechanism, which allows for detection and mitigation of stray points without propagating their influence. However, the paper notes limitations when dealing with wild high-intensity outliers, suggesting the method works best for moderate noise where outliers are not excessively deviant from the underlying surface. The iterative refinement process further enhances these capabilities by progressively correcting point positions and normals, making the method well-suited for point clouds affected by moderate noise commonly encountered in 3D acquisition systems.

The method shows particular advantages in preserving sharp features while denoising, making it suitable for point clouds containing a mix of noise types. The anisotropic neighborhood computation and specialized sharp feature detection allow the algorithm to maintain thin geometric features even

in the presence of Gaussian or moderate impulsive noise. However, the authors acknowledge that the method's performance degrades with extremely high noise intensities where topological issues may arise, and it may require pre-filtering for severe normal outlier contamination. The balanced approach of feature-aware processing and iterative refinement makes this method particularly effective for point clouds, where preserving both sharp and smooth features is critical, though its effectiveness is contingent on noise intensity and outlier severity.

The method demonstrates robust performance against various types of noise encountered in practical applications. For *structured noise* (e.g., sensor-specific artifacts in 3D scanning systems), the anisotropic neighborhood computation adapts to local geometric distortions, suppressing artifacts while preserving sharp features. With *non-uniform noise* (varying intensity across regions), the iterative normal filtering and point-updating steps dynamically adjust smoothing strength based on localized noise estimates. However, it struggles with *topological noise* (e.g., *data gaps*, *scan stitching errors* or severe surface tears), where neighborhood definitions become unreliable.

For *mixed noise* (e.g. Gaussian + sparse outliers), the dual strategy of outlier rejection (via small-neighborhood analysis) and feature-aware smoothing delivers balanced results, though extreme cases may require pre-filtering. The method also addresses *geometric texture noise* (unwanted high-frequency patterns), effectively removing fine surface details while retaining macroscopic shape integrity. Its feature-sensitive design proves particularly effective for point clouds with overlapping noise types, though performance degrades when noise intensities exceed moderate thresholds or disrupt local topology.

3.8. Feature Graph Laplacian Regularization

The Graph Laplacian Regularization method proposed in [13] is particularly effective for removing Laplacian noise due to its ℓ_1-norm fidelity term in the optimization formulation. Unlike Gaussian noise, which is best modeled with the ℓ_2-norm fidelity, Laplacian noise exhibits sparsity with occasional large outliers that are better captured by the ℓ_1-norm. This robustness to outliers allows the method to effectively eliminate Laplacian noise while preserving fine geometric details in the point cloud that might otherwise be oversmoothed by ℓ_2-based approaches. The ℓ_1 formulation provides the mathematical foundation for handling the characteristic of a heavy-tailed Laplacian noise distribution.

A key strength of the method lies in its Signal-Dependent Feature Graph Laplacian Regularizer (SDFGLR), which operates on surface normals rather than point coordinates directly. This approach maintains sharp edges and geometric structures while denoising, as the graph weights adapt according to both surface normals and point positions. For Laplacian noise, which can create localized distortions in the point cloud, this signal-dependent regularization proves crucial for preserving geometric fidelity. The graph-based formulation naturally handles the non-uniform noise distribution typical of Laplacian corruption.

The method employs an Accelerated Proximal Gradient (APG) algorithm to solve the non-differentiable ℓ_1-norm optimization problem efficiently. APG's convergence properties make it well-suited for Laplacian noise removal, as it can handle the non-smooth objective function while maintaining computational stability. Furthermore, the method's adaptive noise variance estimation automatically adjusts the regularization parameter γ for different Laplacian noise levels, eliminating the need for manual tuning. This adaptability ensures consistent performance across varying noise intensities, while the graph Laplacian framework provides the geometric constraints needed for high-quality denoising.

3.9. Fine-Granularity Dynamic Graph Convolutional

The GD-GCN method [49] addresses motion noise disambiguation by explicitly modeling temporal correspondences in non-static scans. Unlike conventional static denoising approaches, GD-GCN frames point cloud processing as a continuous spatio-temporal system through its Micro-Step Temporal Graph Convolution (MST-GConv) module. This formulation as a neural ordinary differential

equation (ODE) enables the network to learn infinitesimal trajectory adjustments, effectively separating legitimate motion from sensor-induced noise artifacts.

This method performs well at reconstructing continuous surfaces from point clouds corrupted by *intrinsic-parametric* noise (particularly Gaussian/uniform) through its Riemannian metric learning and micro-step temporal convolutions. Its spectrum-adaptive filters show the strongest performance against *systemic-structured* sensor noise, where dynamic graph topology adaptation captures depth-dependent artifacts. While effective for measurement noise respecting manifold assumptions, the method faces limitations with *non-parametric* outliers (e.g., impulsive noise) and *extrinsic* interference like specular reflections where geometric consistency breaks down. The fusion of differential geometry with learnable spectral filters makes it particularly suitable for surface-aware denoising in acquisition processes with known noise characteristics.

Noise robustness: The method demonstrates particular effectiveness against *scene-induces interferences* categories, especially those involving temporal inconsistencies and exhibiting spatio-temporal dependencies in dynamic acquisitions such as misaligned LiDAR scans of moving objects or dynamic obstructions. GD-GCN's geometric graph construction employs Riemannian metrics to preserve intrinsic manifold structures across successive frames, allowing simultaneous adaptation to both local shape deformations and global motion patterns. The Bernstein polynomial-based spectral filtering component provides additional stability by bounding eigenvalue perturbations, a crucial feature for maintaining geometric fidelity while processing temporally evolving point clouds.

Main advantages: GD-GCN's primary strength lies in its handling of dynamic noise through the MST-GConv's ODE formulation, which enables iterative refinement of point positions across pseudo-temporal steps. This approach shows remarkable robustness against non-rigid motion artifacts that typically challenge conventional denoising methods. The Riemannian graph construction further enhances performance by maintaining geodesic consistency, avoiding the distortion effects common in Euclidean-based graph convolutions. Additionally, the spectral filtering mechanism ensures stable operations even with irregular sampling patterns, a critical requirement for real-world LiDAR applications.

GD-GCN's continuous convolution framework represents a significant advancement over discrete-step methods, providing nuanced control over noise removal while preserving legitimate motion trajectories; a capability that addresses several longstanding challenges in processing non-static 3D data acquisitions. However, the iterative nature of the ODE solver introduces higher computational costs compared to static denoising approaches, though this trade-off generally results in superior denoising fidelity.

3.10. Second-Order Regularization and Bi-Tensor Voting

In [23], the authors present a second-order normal filtering method that demonstrates distinct performance between different types of noise in point clouds. The technique stands out particularly against *intrinsic-parametric* noise (e.g. uniform distributions) through its anisotropic second-order operator, which preserves sharp features while smoothing continuous surfaces. The bi-tensor voting scheme shows robustness against *systemic-structured* noise such as scan artifacts and misalignments by effectively distinguishing genuine geometric features from systematic perturbations.

The method achieves optimal performance for surface reconstruction affected by *discrete noise* and *outliers*, where its RANSAC-based multi-normal estimation reliably identifies underlying planes amidst noisy samples. However, it faces limitations with *non-parametric* noise types like specular reflections or severe ambient interference, where geometric continuity assumptions break down. The computational stability of the augmented Lagrangian solver enhances practical applicability, though the multi-normal processing increases complexity for feature-rich scenes.

In essence, it merges point and normal tensor voting to distinguish real geometry from noise. Although designed for static denoising, its noise-resistant approach could help filter dynamic obstructions without distorting surfaces. The method suppresses artifacts similarly to how pseudo-features occur in dynamic clusters, suggesting potential for temporal adaptation. As its key strength, the

approach outperforms in preserving sharp edges while filtering *structured sensor noise*, effectively combining second-order regularization with tensor analysis. This makes it particularly valuable for high-precision scanning applications where geometric fidelity is critical, though performance declines in dynamic environments with temporal occlusions.

3.11. Total Denoising Unsupervised Learning

The unsupervised denoising algorithm proposed in [50] addresses the challenge of *total noise* in 3D point clouds (deviations in all coordinates, without a reliable regular grid), where both spatial coordinates (domain) and attributes (range) are corrupted.

The method excels against *statistical noise* models, particularly parametric distributions like Gaussian noise or systematic sensor bias (e.g., LiDAR artifacts), by leveraging a *spatial-appearance prior* to steer convergence toward the closest clean point on a noisy manifold. This prior resolves the ambiguity of *total noise*, where a corrupted observation could originate from multiple clean points, by enforcing proximity in both geometry and (optionally) color space. The algorithm outperforms supervised methods for *independent, low-to-medium intensity noise* due to its ability to generalize directly from noisy data without clean references. The key strength of the method lies in its unsupervised adaptability to *intrinsic parametric noise* while preserving sharp features.

However, its performance diminishes for *structured noise* (e.g., occlusions, reconstruction artifacts) or *non-parametric outliers*, as the unimodal prior struggles to disambiguate complex corruptions. Notably, it fails to address *global noise* (e.g., rigid displacements) or *scene-dependent interferences*, as these violate its core assumption of local deviations around a clean surface. The method remains limited by its reliance on spatial locality and unimodality, excluding broader noise classes in real-world scans.

Topology-aware Progressive Noising and Denoising of Point Clouds towards Upsampling

3.12. Topology-aware Progressive Noising and Denoising towards Upsampling

The TP-NoDe method [36] demonstrates particular effectiveness in addressing data gaps in point clouds, a common imperfection that arises during 3D acquisition or reconstruction. By employing density-aware k-nearest neighbors (DA-kNN) to extract local neighborhoods and strategically introducing topological noise (perturbations that respect the underlying geometric structure rather than purely random displacements), the method can intelligently fill missing regions while preserving the shape's fidelity. This approach outperforms traditional upsampling methods that often create artificial geometries or fail to maintain uniform point distributions in gapped areas.

For non-uniform noise patterns, the TP-NoDe progressive noising and denoising framework shows significant advantages. The density-aware neighborhood selection adapts to varying noise distributions across the point cloud, unlike fixed-k NN approaches that struggle with spatially varying noise characteristics. The method's ability to handle different *statistical noise types* (Gaussian, Laplacian, Discrete, uniform, and Covariance noise) through its perturbation model makes it robust against the non-uniform noise commonly encountered in real-world 3D scanning scenarios.

A distinctive feature of TP-NoDe is its specialized handling of topological noise (understood as structured perturbations guided by local surface properties rather than purely statistical distributions) through the DA-kNN algorithm. By dynamically adjusting the neighborhood size based on local density, the method better preserves sharp features and intricate geometric details that are often corrupted by conventional denoising approaches. This makes it particularly suitable for point clouds that contain both *structured noise patterns* and complex geometries, where maintaining topological fidelity is crucial for accurate upsampling.

3.13. Handling Long-Range and Discrete Outliers in Surface Reconstruction

Recent advances in surface reconstruction have developed specialized approaches for handling both long-range clustered outliers and discrete scattered outliers. As discussed in [19], the mono-oriented grouping method [51] provides an effective solution for long-range outliers by organizing point clouds into octrees and clustering through active contours. This approach identifies outlier clus-

ters when bounded by neighboring mono-oriented groups with identical orientation, demonstrating particular effectiveness against structured noise and distant outlier clusters. The method's integration with implicit surface reconstruction preserves features while handling incomplete data, though its performance depends heavily on octree resolution and incurs higher computational costs than local statistical methods.

For discrete outliers, the spectral reconstruction approach [52] employs graph partitioning techniques through Delaunay tetrahedralization and two-stage spectral analysis of the dual Voronoi diagram. By constructing a weighted pole graph and partitioning via eigenvector analysis, the method automatically filters discrete outliers while maintaining watertight surfaces. Both approaches share a global analysis perspective (the mono-oriented method through orientation voting and the spectral method through eigenvector partitioning) which provides superior robustness against non-local noise patterns compared to local statistical methods.

Although these methods excel in their respective domains (long-range clustered outliers and discrete scattered outliers), they share common limitations with Gaussian noise distributions where smoothing techniques may prove more effective. The mono-oriented approach demonstrates particular strength with scanning artifacts and structured noise, while the spectral method shows advantages for organic shapes where topological completeness is prioritized. Both approaches represent significant advances in handling challenging outlier cases that traditional local methods struggle with, albeit at higher computational costs. Their complementary strengths suggest potential for hybrid approaches in future work, particularly for datasets containing both types of outlier patterns.

3.14. Differentiable Manifold Reconstruction

The DMR method [53] is particularly effective in handling Gaussian noise and impulsive noise/outliers, as highlighted in the experimental results. The differentiable pooling operator learns to identify and retain points closer to the underlying surface, filtering out high-frequency Gaussian noise by adaptively downsampling the point cloud. For impulsive noise, the pre-filtering step and manifold reconstruction explicitly mitigate the impact of outliers by leveraging local and non-local feature representations, which are robust to sparse, large-magnitude perturbations. The method's ability to reconstruct the underlying manifold from subsampled points makes it well-suited for these statistical noise models.

The method also performs well for *scene-induced interferences* such as *specular/reflective* surfaces and *dynamic obstructions*, as these often introduce structured noise that corrupts local geometry. By learning patch manifolds from neighborhood features, DMR can implicitly compensate for distortions caused by reflections or transient objects, as seen in experiments on the real-world dataset Paris-rue-Madame [54]. The adaptive pooling operation further reduces the influence of such interferences by prioritizing points with stable, low-noise features. This contrasts with traditional methods that rely on fixed filtering rules and struggle with complex scene-induced artifacts.

The manifold reconstruction decoder infers missing or corrupted geometry by resampling from learned local surfaces, effectively filling gaps while preserving sharp features. The paper demonstrates strong performance on real-world LiDAR noise (an unseen structured noise pattern), indicating generalization to sensor-specific artifacts. However, the method may be less suitable for uniform noise or unidirectional noise, as these can uniformly perturb the point cloud, making it harder for the pooling layer to distinguish clean points without explicit prior knowledge of the noise direction or distribution.

3.15. Self-Supervised Deep Depth Denoising

The self-supervised denoising method proposed in [37] demonstrates particular effectiveness against various types of noise encountered in depth sensing. For structured sensor noise such as the characteristic patterns from Intel RealSense D415 cameras, the combination of partial convolutions and robust loss functions of the method prevents the boundary spraying artifacts that plague other learning-based approaches like DDRNet [55]. The architecture handles both the high-frequency noise

components and low-frequency distortion patterns typical of consumer depth sensors, outperforming traditional bilateral filters while avoiding the over-smoothing tendencies of methods like DRR that were trained on planar-dominated datasets. The approach shows special resilience to the mixed noise profiles found in real sensor data, where *mechanical instability* and *sensor-specific* noise often appear simultaneously.

In complex acquisition scenarios that involve multiple sensors or dynamic scenes, the method maintains robust performance when others, such as RGF [56], fail. The experiments with four-camera Poisson reconstruction setups reveal the method's ability to handle *stitching artifacts* and inconsistent depth measurements that create "balloon" distortions in the fused output. This self-supervised approach automatically adapts to varying noise conditions in different sensor viewpoints. The KinectFusion experiments further demonstrate how the method preserves geometric detail while suppressing the flying pixels and temporal instability that usually cause tracking failures.

3.16. Variational Formulations for Noise Types

As demonstrated in [57], variational methods have proven highly effective in removing noise removal in the grid domain. These approaches combine two key components: a data fidelity term that maintains consistency with observed data, and a regularization term that controls solution smoothness while preserving important features. The method's effectiveness stems from its adaptability to different noise statistical models and geometric structures.

The choice of data fidelity term depends on the specific noise properties. For Gaussian noise common in sensor data, minimizing squared differences between processed and observed data works best. Poisson noise requires terms combining observations with logarithmic transformations, while Laplace noise benefits from absolute difference measures. Impulse noise like salt-and-pepper artifacts is best handled by counting non-zero discrepancies.

Regularization strategies vary according to the desired geometric properties. Isotropic total variation promotes general smoothness while respecting edges, whereas anisotropic versions better preserve sharp features. Recent sparsity-promoting regularizers that track non-zero gradient entries have shown particular success in maintaining fine details.

For 3D point clouds, these methods adapt well by replacing image gradients with spatial differences between points or normal variations, for example. The approach effectively removes noise while preserving crucial geometric features, making it particularly valuable for point cloud processing across different applications.

3.17. CNN-based LiDAR in Adverse Weather

The CNN-based WeatherNet denoising architecture [28] demonstrates strong effectiveness against scene-induced statistical noise, such as speckle noise and Poisson-distributed interference from fog and rain particles. These noise types are well-handled because the CNN learns their intensity-dependent patterns and spatial correlations, unlike traditional filters that rely solely on geometric heuristics. Unlike traditional geometric filters (e.g., DROR [58] or ROR [59]), which struggle with dense scattering patterns, the learned feature representations in WeatherNet distinguish between valid surfaces and atmospheric noise by analyzing both spatial distribution and intensity characteristics. By modeling non-linear intensity distributions of weather-induced reflections and preserving structural details through dilated convolutions, the method achieves robust performance in adverse conditions.

For dynamic scene interferences—including *specular reflections* and adverse weather effects—the method significantly outperforms geometric filtering approaches. The 2D cylindrical projection maintains critical structural relationships for identifying weather patterns, while the modified LiLaNet architecture [60], enhanced with dropout and dilated convolutions, adapts to varying point densities caused by environmental conditions. Notably, the method retains detection capability for dynamic objects such as pedestrians and cyclists in heavy fog, where conventional filters fail, making it well-suited for real-world mobile applications. Inclusion of intensity-based data augmentation further improves generalization to natural rainfall patterns.

However, the method exhibits limitations when dealing with sensor-specific structured noise and unidirectional errors. While the autolabeling procedure mitigates minor measurement inconsistencies, the CNN's effectiveness depends on the training data coverage of specific sensor artifacts. Although the architecture supports real-time processing, scan stitching errors and mechanical vibrations may require additional temporal filtering, which is not addressed in the current framework.

3.18. Edge-guided Second-order TGV for Depthmap

The ESTGV method [61] outperforms in handling high-variance Gaussian noise due to its variational formulation, which minimizes an energy functional combining a regularization term and a data fidelity term. The model's key strength lies in its adaptive edge preservation via a Canny-based indicator function, which selectively weakens diffusion in high-gradient edge regions while enforcing stronger smoothing in low-gradient homogeneous areas. This makes ESTGV particularly effective for high-intensity Gaussian noise, where traditional PDE-based or filtering methods often fail to balance noise removal and edge retention.

ESTGV is optimally suited for additive *statistical noise* due to its L^2-norm fidelity term, which aligns with Gaussian-distributed errors. The method's second-order TGV regularization also mitigates staircasing artifacts common in TV-based approaches, making it theoretically robust against moderate speckle and uniform noise. However, its performance may degrade for impulsive noise or scene-induced interferences such as specular surfaces, as these violate the Gaussian assumption and require specialized fidelity terms.

Although ESTGV outperforms benchmarks like WNNM [62] for Gaussian noise, its reliance on gradient-based edge detection makes it less effective for structured noise or data gaps. The method assumes noise homogeneity, so *unidirectional* noise or *dynamic obstructions* may cause uneven denoising. Its iterative first-order primal-dual solver is convergent and computationally efficient compared to deep learning approaches. Although here the focus is on regular depth maps, which are markedly different from unstructured 3D point clouds, it is nevertheless worth mentioning these considerations.

3.19. Regularization of a Statistical Low-Dimensional Manifold

The proposed R-SLDM method in [63] is particularly effective for handling two main types of noise in 3D point clouds: Gaussian noise and Laplacian noise. The method takes advantage of the inherent low-dimensional manifold structure of point clouds to regularize the denoising process, making it robust against these statistical noise types.

Gaussian noise, characterized by its symmetric distribution around zero, is well-addressed by R-SLDM due to its statistical modeling approach. The method minimizes the impact of this noise by optimizing the fidelity term in the objective function, which ensures that the denoised point cloud remains close to the observed data while reducing the noise variance. The regularization term further enhances this by preserving the geometric structure of the underlying manifold, effectively smoothing out the Gaussian noise without oversmoothing the salient features of the point cloud.

Laplacian noise, which is more impulsive and sparse compared to Gaussian noise, is also effectively handled by R-SLDM. The method identifies and removes outliers by statistically analyzing local neighborhood distances, setting a threshold to detect and eliminate points that deviate significantly from the manifold structure. This approach is particularly suitable, as it targets the sparse, large-magnitude perturbations that are characteristic of this noise type. The alternating optimization framework ensures that both the fidelity and regularization terms are balanced, leading to effective noise removal while maintaining the geometric integrity of the point cloud.

Nevertheless, this method is not intended for other types of noise explicitly described in the taxonomy. So its main strength lies in handling the combined presence of Gaussian and Laplacian noise, which are common in practical scenarios. However, it may fail when dealing with more complex, scene-dependent, or uncontrolled scans.

3.20. TripleMixer in Adverse Weather

The TripleMixer [24] algorithm presents a novel multi-perspective approach to point cloud denoising, particularly effective against *extrinsic-structured* noise types such as adverse weather conditions (snow, fog, and rain) that dominate real-world LiDAR acquisitions. Its three-layer architecture demonstrates unique advantages against different noise categories according to our taxonomy:

The Geometry Mixer (GMX) Layer excels at handling *spatially structured* noise through voxel-wise downsampling and attentive pooling of local neighborhoods, preserving geometric fidelity while suppressing weather-induced outliers. This makes it particularly suitable for *environmental interference* noise (e.g., snow and rain particles) that exhibit spatial clustering patterns.

The Frequency Mixer (FMX) Layer shows exceptional performance against *global weather patterns*, especially dense fog (its best-performing case), through multi-scale wavelet analysis. This layer effectively separates the characteristic low-frequency spectra of fog from true surface geometries in the frequency domain, addressing the *non-parametric* nature of atmospheric scattering effects.

For *systemic noise*, the Channel Mixer (CMX) Layer provides robustness against sensor-specific artifacts through cross-channel feature integration. This capability proves valuable for *structured sensor noise* that manifests differently across acquisition channels (e.g., LiDAR intensity vs. spatial coordinates).

While demonstrating comprehensive coverage, TripleMixer shows relative limitations with *intrinsic-parametric* noise (e.g., Gaussian, Poisson, etc.), where simpler statistical models might suffice, and requires additional temporal processing for *dynamic obstructions*. The voxel-based approach may also underperform with *specular surface noise* requiring sub-voxel precision. Nevertheless, its hybrid geometric-spectral approach sets a new standard for weather-robust 3D perception, particularly in autonomous driving scenarios where complex noise mixtures prevail.

As noted in [24], unlike statistical methods, deep learning excels at directly learning weather-induced noise patterns in point clouds. Although general-purpose segmentation networks [64–66] have been adapted for denoising, they lack specialization. Existing denoising networks fall into two categories: *range-image-based* and *raw 3D point cloud-based*. WeatherNet [28] modifies LiLaBlock [60] to filter weather noise, while 4DenoiseNet [67] leverages spatiotemporal KNN convolutions and attention, but requires temporally continuous LiDAR scans. LiSnowNet [68] combines FFT and CNNs for unsupervised snow removal. However, range-image methods suffer from projection artifacts and spatial information loss. In contrast, 3D-OutDet [69] processes raw 3D points using Neighborhood (NH) Convolution, specifically targeting snowflake removal without structural degradation.

3.21. Dynamic Statistical Outlier Removal

Several statistical approaches have been developed for LiDAR denoising in *adverse weather* and *light interference* noise. The basic Statistical Outlier Removal (SOR) [59] and Radius Outlier Removal (ROR) [59] filters were enhanced by dynamic variants such as Dynamic ROR (DROR) [58] and Low-Intensity Outlier Removal (LIOR) [70]. More advanced methods include Dynamic Distance-Intensity Outlier Removal (DDIOR) [71] and Conditional Random Fields Outlier Removal (CRFOR) [72], which incorporate additional sensor data and temporal information.

The Dynamic Statistical Outlier Removal (DSOR) filter [73] provides superior performance by dynamically adjusting its parameters based on point distance from the sensor. This approach effectively handles LiDAR's natural density variations while maintaining computational efficiency, making it particularly suitable for real-time applications in severe weather conditions.

3.22. Dynamic Occlusion in TLS Point Clouds Sequence

The dynamic occlusion detection and inpainting method [22] demonstrates particular effectiveness against *extrinsic-structured* noise types, especially *light interference*, *vibrations or mechanical stability*, and *dynamic obstructions* such as pedestrians in TLS scans. Its ray-voxel background modeling approach combines geometric and intensity information to robustly distinguish persistent scene structures

from moving objects, achieving optimal performance for *dynamic obstructions* that create temporal inconsistencies in scan sequences. The adaptive logistic function weighting mechanism proves crucial for handling cases where either geometry or intensity data alone would fail due to camouflage effects between foreground and background elements.

The method shows strong capabilities for *scene-induced interferences* including both slow-moving crowds (e.g., historical site) and fast-moving athletes (e.g., gymnasium), with a GMM-based approach successfully rejecting irregularly shaped and incomplete occlusions. However, it faces limitations with *specular reflections* at foreground-background junctions, where mirrored points can initially corrupt the background model until sufficient scans are acquired. The technique's multi-scan requirement (minimum 3, optimal more than 6 scans) makes it less suitable for single-scan applications or environments with extremely rapid scene changes.

Key strengths include the method's precision in background point extraction and its ability to preserve complex static structures while removing dynamic elements. The adaptive logistic function weighting mechanism ensures robustness against geometry/intensity camouflage, outperforming fixed-weight or single-feature approaches.

3.23. Dynamic Graph CNN

The method proposed in [16] is competent in handling statistical noise models with spatial coherence, particularly Gaussian noise and uniform noise introduced during acquisition. The initial gradient estimation unit leverages local neighborhood features through DGCNN layers, making it robust to moderate Gaussian perturbations where noise follows $\mathcal{N}(0, \sigma^2)$. The temporal gradient field construction further suppresses such noise by averaging transformed gradients from adjacent frames, effectively exploiting the noise's zero-mean property. This aligns with the method's assumption of additive noise, where the gradient potential energy minimizes deviations from the underlying surface.

For scene-induced interferences like dynamic obstructions or specular reflections, the rigid motion-based correspondence search proves advantageous. By modeling patch transformations through classical mechanics, the method distinguishes between noise and legitimate motion artifacts. However, the inertia matrix formulation assumes a uniform mass distribution, making it less ideal for impulsive noise/outliers that violate this assumption. Even so, the results confirm its superior performance against Gaussian noise and simulated LiDAR data as well as its robustness to outliers.

The framework shows reduced efficacy for acquisition imperfections like structured noise or data gaps. While the temporal correspondence search (Algorithm 2) can interpolate small gaps via k-NN graphs, the gradient ascent struggles with sensor-specific patterns (e.g., stripe noise in LiDAR). The ablation study confirms this: it degrades significantly with high-variance Gaussian noise, as the method primarily optimizes for minimization of the ℓ_2-norm minimization rather than structural integrity metrics.

3.24. PD-Flow: Denoising Framework with Normalizing Flows

PD-Flow [74] demonstrates superior denoising performance across various noise types, particularly excelling against broad *statistical noise* types, such as Gaussian noise (isotropic and non-isotropic), uniform noise distributions, unidirectional noise, and *discrete noise*. Its effectiveness stems from a novel flow-based architecture employing Normalizing Flows (NFs) to learn bijective mappings between noisy and clean point clouds, preserving geometric structures while filtering perturbations. The method's invertible NF framework maintains geometric fidelity by learning a disentangled latent space that separates noise components from underlying surface features, outperforming both traditional and deep learning baselines.

A critical innovation is the Feature-Based Masking (FBM) filter, which explicitly decomposes the latent space into clean and noisy channels. This architecture proves particularly effective for non-isotropic and *sensor-specific* noise like LiDAR artifacts, where it suppresses directional distortions while preserving surface topology. The FBM mechanism enables selective noise suppression without corrupting geometric details, while the flow-based formulation inherently promotes point distribution

uniformity. These properties make PD-Flow robust against real-world scanning noise where irregular sampling and non-uniform perturbations are common challenges.

The method exhibits remarkable generalization to unseen noise types (*unidirectional*, *discrete*) without retraining, attributable to its probabilistic modeling of clean point cloud distributions rather than noise-specific patterns. The disentangled latent representation allows adaptive filtering of diverse perturbations, including sparse outliers and structured artifacts, while maintaining surface smoothness. This adaptability is validated through extensive tests on varied noise configurations, demonstrating consistent performance where traditional methods require parameter tuning for each noise type.

PD-Flow's combination of invertible transformations, latent space disentanglement, and distribution-aware processing makes it uniquely suited for practical applications with unpredictable or *mixed noise* characteristics. The architecture's ability to handle both synthetic and real-world noise while generalizing to novel perturbations addresses a key limitation in current denoising pipelines.

3.25. Spatial-Temporal Graph Learning for 3D Dynamic Denoising

The proposed spatio-temporal graph learning method in [25] is particularly effective against *statistical additive noise* due to its Mahalanobis distance-based edge weighting and Graph Laplacian Regularizer minimization. This noise type aligns well with the method's assumption of smooth signal variation across connected nodes in the graph. For example, the exponential kernel in edge weight computation naturally suppresses Gaussian-distributed perturbations, while the trace constraint on the metric matrix prevents overfitting to Laplacian or uniform noise outliers. The method's iterative proximal gradient optimization further enhances robustness against these additive noise models.

For *specular/reflective surfaces* and *light interference*, the method shows strong performance through its surface normal-aware feature vector and patch-based similarity metric. The inclusion of angular differences between normals in the feature space helps to distinguish true geometry from transient lighting artifacts. However, the method is less suited for dynamic obstruction effects that violate the temporal consistency assumption, as these may introduce structured noise that corrupts the inter-frame correspondence estimation.

The algorithm excels at addressing *structured/sensor-specific noise* and *scan stitching errors* through its joint spatial-temporal optimization framework. The alternating minimization between point cloud updates and graph learning adaptively corrects sensor-specific distortions while preserving genuine geometric features. However, the patch-based approach may struggle with *data gaps* that exceed the local search window size, as missing connectivity information cannot be reliably inferred from graph topology alone. The method's performance degrades gracefully with increasing noise variance, maintaining superiority over static denoising approaches even at high noise levels.

3.26. Mixed Noise

The work of [75] proposes a score-based method to address mixed noise in 2D images, focusing on multiplicative Rayleigh noise combined with additive Gaussian noise. Although centered on images, their approach offers valuable insights for 3D point cloud denoising. A key challenge is the lack of a closed-form solution for the joint distribution $p(y|x)$, which represents the probability of observing noisy data y given the clean data x; this requires approximations that assume additive noise dominance, an assumption often invalid for 3D sensors like LiDAR where noise depends on the signal.

Their iterative optimization method is effective but computationally expensive, limiting scalability to large 3D datasets. Unsupervised and self-supervised methods such as Noise2Score [76] and Neighbor2Neighbor [77] also face challenges with non-exponential and spatially correlated noise, respectively, which are common in 3D data. The score-based approach shows promise but requires precise noise parameter tuning, difficult in 3D due to unknown noise models.

The treatment of multiplicative Rayleigh noise and the corresponding iterative solver could be adapted for depth-dependent noise in 3D sensors. Their framework for mixed noise, combining multiplicative and additive components, may inspire similar strategies for 3D data corrupted by

speckle and thermal noise. However, reliance on known noise parameters restricts its applicability to real-world scans.

In a related manner, [14] shows that Markov–Middleton models capture bursty noise dependencies through state transitions, a concept directly applicable to 3D point clouds when outliers cluster spatially (e.g., due to persistent scene obstructions and impulsive noise). For bursty noise, memory-aware approaches—such as spatiotemporal filtering or graph-based smoothing—outperform pointwise filters by leveraging contextual coherence. While the memoryless Bernoulli model is sufficient for isolated outliers, Markovian extensions are critical for bursty artifacts, which are common in dynamic captures or multi-sensor fusion. The tractability of the Bernoulli–Gaussian mixture also makes it a promising candidate for hybrid noise modeling in 3D, where outliers often coexist with structured noise (e.g., Gaussian jitter).

3.27. Cross-Category Effectiveness Matrix

The Cross-Category Effectiveness Matrix compares the performance of various filtering methods across different types of noise—statistical, scene-induced, and acquisition-related.

Table 1. Point Cloud Denoising Effectiveness by Noise Category

Method	Gauss	Laplace	Uniform	Impulse	Scene-Induced	Acquisition
GGPSR [16]	H	H	M	M	M	H
GPDNet [41]	H	M	M	L	L	H
SBPCD [2]	H	M	M	L	L	M
PCN [40]	H	L	M	M	L	M
TGV [42]	H	M	H	L	M	H
ANSD [48]	H	M	M	M	M	H
FGLR [13]	M	H	L	L	L	M
GD-GCN [49]	H	M	H	L	H	H
TDUL [50]	H	M	M	L	L	M
TP-NoDe [36]	H	H	H	M	M	H
TMX [24]	M	M	M	H	H	H
PD-Flow [74]	H	H	H	M	M	H

Effectiveness: H=High, M=Medium, L=Low

4. Conclusions

The interdependence of noise sources demands hybrid solutions that jointly consider statistics, physics, and learning. Our taxonomy highlights that treating noise dimensions in isolation leads to suboptimal or brittle methods, especially in realistic imaging pipelines where mixed, scene-dependent, and sensor-specific artifacts coexist.

Call to action: Open-source, reproducible toolkits that address all three taxonomy axes, integrating classical models, physically grounded simulation, and data-driven components under a common, extensible framework.

4.1. Critical Problems Analysis

Unresolved challenges:

- No unified framework for mixed noise (e.g., Gaussian + impulsive) that is both theoretically sound and practically deployable.
- Limited real-time performance for dynamic or large scenes, particularly when leveraging complex learned or physics-based priors.
- Sensor-specific biases in learning-based denoising that hinder cross-device generalization and long-term robustness.

- Lack of standardized evaluation protocols for hybrid methods combining model-based and data-driven components.
- Insufficient interpretability and uncertainty quantification in state-of-the-art denoisers.

4.2. Future Work and Future Directions

- Physics-augmented neural networks for noise disentanglement, jointly learning sensor models and scene priors.
- Benchmark datasets with complex ground-truth noise annotations, covering mixed regimes, temporal dynamics, and multiple sensor types.
- Modular open-source pipelines that couple forward noise simulation with inverse denoising, enabling end-to-end evaluation.
- Future research should prioritize generative restoration over simple filtering, such as 3D Diffusion Models, which treat denoising as a reverse stochastic process to recover the underlying manifold, leveraging latent diffusion to reconstruct clean surfaces in extreme low-density or high-noise scenarios where traditional geometric heuristics fail.
- Domain adaptation and continuous learning strategies to reduce sensor- and data set-specific biases.
- Real-time hardware-aware algorithms optimized for edge devices and embedded vision systems.

Author Contributions: The sole author contributed to the conceptualization, design, analysis, and writing of this manuscript.

Funding: Not applicable.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available from the corresponding authors on request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Poux, F. The Smart Point Cloud: Structuring 3D intelligent point data. PhD thesis, 2019. https://doi.org/10.13140/RG.2.2.20457.75367.
2. Luo, S.; Hu, W. Score-Based Point Cloud Denoising, 2021.
3. Park, J.H.; Shin, Y.D.; Bae, J.H.; Baeg, M.H. Spatial Uncertainty Model for Visual Features Using a Kinect™ Sensor. *Sensors* **2012**, *12*, 8640–8662. https://doi.org/10.3390/s120708640.
4. Korkalo, O.; Takala, T. Measurement Noise Model for Depth Camera-Based People Tracking. *Sensors* **2021**, *21*. https://doi.org/10.3390/s21134488.
5. Zhou, L.; Sun, G.; Li, Y.; Li, W.; su, Z. Point cloud denoising review: from classical to deep learning-based approaches. *Graphical Models* **2022**, *121*, 101140. https://doi.org/10.1016/j.gmod.2022.101140.
6. Chen, C.; Xiong, Z.; Tian, X.; Zha, Z.J.; Wu, F. Real-World Image Denoising with Deep Boosting. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **2020**, *42*, 3071–3087. https://doi.org/10.1109/TPAMI.2019.2921548.
7. Niknejad, M.; Figueiredo, M. Poisson Image Denoising Using Best Linear Prediction: A Post-processing Framework **2018**. https://doi.org/10.48550/arXiv.1803.00389.
8. Chowdhury, M.; Zhang, J.; Qin, J.; Lou, Y. Poisson image denoising based on fractional-order total variation. *Inverse Problems Imaging* **2020**, *14*, 77–96. https://doi.org/10.3934/ipi.2019064.
9. Chukhovskii, F.N.; Konarev, P.V.; Volkov, V.V. Denoising of the Poisson-Noise Statistics 2D Image Patterns in the Computer X-ray Diffraction Tomography. *Crystals* **2023**, *13*. https://doi.org/10.3390/cryst13040561.
10. Rodrigues, I.; Sanches, J.; Bioucas-Dias, J. Denoising of medical images corrupted by Poisson noise. In Proceedings of the 2008 15th IEEE International Conference on Image Processing, 2008, pp. 1756–1759. https://doi.org/10.1109/ICIP.2008.4712115.

11. Bodduna, K.; Weickert, J. Poisson Noise Removal Using Multi-Frame 3D Block Matching, 2019, [arXiv:eess.IV/1909.08281].
12. Deschaud, J.E.; Goulette, F. POINT CLOUD NON LOCAL DENOISING USING LOCAL SURFACE DESCRIPTOR SIMILARITY.
13. Dinesh, C.; Cheung, G.; Bajić, I.V. Point Cloud Denoising via Feature Graph Laplacian Regularization. *IEEE Transactions on Image Processing* **2020**, *29*, 4143–4158. https://doi.org/10.1109/TIP.2020.2969052.
14. Shongwe, T.; Vinck, A.J.H.; Ferreira, H.C. A Study on Impulse Noise and Its Models. *SAIEE Africa Research Journal* **2015**, *106*, 119–131. https://doi.org/10.23919/SAIEE.2015.8531938.
15. Roos-Hoefgeest, S.; Roos-Hoefgeest, M.; Álvarez, I.; González, R.C. Simulation of Laser Profilometer Measurements in the Presence of Speckle Using Perlin Noise. *Sensors* **2023**, *23*. https://doi.org/10.3390/s23177624.
16. Chen, H.; Du, B.; Luo, S.; Hu, W. Deep Point Set Resampling via Gradient Fields. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **2021**, *45*, 2913–2930.
17. Vlasenko, A.; Schnorr, C. Physically Consistent and Efficient Variational Denoising of Image Fluid Flow Estimates. *IEEE Transactions on Image Processing* **2010**, *19*, 586–595. https://doi.org/10.1109/TIP.2009.2036673.
18. Clason, C. L-infinity fitting for inverse problems with uniform noise. *Inverse Problems* **2012**, *28*, 104007. https://doi.org/10.1088/0266-5611/28/10/104007.
19. Chen, H.; Shen, J. Denoising of point cloud data for computer-aided design, engineering, and manufacturing. *Engineering with Computers* **2018**, *34*, 523–541. https://doi.org/10.1007/s00366-017-0556-4.
20. Berger, M.; Tagliasacchi, A.; Seversky, L.; Alliez, P.; Levine, J.; Sharf, A.; Silva, C. State of the Art in Surface Reconstruction from Point Clouds. *Eurographics 2014-State of the Art Reports* **2014**.
21. Zhou, Z.; Lu, Q.; Wang, Z.; Huang, H. Detection of Micro-Defects on Irregular Reflective Surfaces Based on Improved Faster R-CNN. *Sensors* **2019**, *19*. https://doi.org/10.3390/s19225000.
22. Chen, C.; Yang, B. Dynamic occlusion detection and inpainting of in situ captured terrestrial laser scanning point clouds sequence. *ISPRS Journal of Photogrammetry and Remote Sensing* **2016**, *119*, 90–107. https://doi.org/https://doi.org/10.1016/j.isprsjprs.2016.05.007.
23. Liu, Z.; Xiao, X.; Zhong, S.; Wang, W.; Li, Y.; Zhang, L.; Xie, Z. A feature-preserving framework for point cloud denoising. *Computer-Aided Design* **2020**, *127*, 102857. https://doi.org/https://doi.org/10.1016/j.cad.2020.102857.
24. Zhao, X.; Wen, C.; Wang, Y.; Bai, H.; Dou, W. TripleMixer: A 3D Point Cloud Denoising Model for Adverse Weather, 2024, [arXiv:cs.CV/2408.13802].
25. Hu, W.; Hu, Q.; Wang, Z.; Gao, X. Dynamic Point Cloud Denoising via Manifold-to-Manifold Distance **2020**. https://doi.org/10.1109/TIP.2021.3092826.
26. Zaman, F.; Wong, Y.P.; Ng, B.Y. Density-based Denoising of Point Cloud, 2016, [arXiv:cs.CV/1602.05312].
27. Ding, S.; Chen, X.; Ai, C.; et al. A noise-reduction algorithm for raw 3D point cloud data of asphalt pavement surface texture. *Scientific Reports* **2024**, *14*, 16633. https://doi.org/10.1038/s41598-024-65233-8.
28. Heinzler, R.; Piewak, F.; Schindler, P.; Stork, W. CNN-Based Lidar Point Cloud De-Noising in Adverse Weather. *IEEE Robotics and Automation Letters* **2020**, *5*, 2514–2521. https://doi.org/10.1109/LRA.2020.2972865.
29. Ji, H.; Zhang, Y.; Chen, S.; Chen, H.; Guo, P.; Zhao, L.; Liang, T.; Zheng, S. Enhanced inversion accuracy of aerosol backscatter coefficient based on simple de-noising operation. *Journal of Quantitative Spectroscopy and Radiative Transfer* **2020**, *257*, 107358. https://doi.org/https://doi.org/10.1016/j.jqsrt.2020.107358.
30. Pomerleau, F.; Breitenmoser, A.; Liu, M.; Colas, F.; Siegwart, R. Noise characterization of depth sensors for surface inspections. In Proceedings of the 2012 2nd International Conference on Applied Robotics for the Power Industry (CARPI), 2012, pp. 16–21. https://doi.org/10.1109/CARPI.2012.6473358.
31. Pistilli, F. Learning Graph-Convolutional Representations for Point Cloud Denoising, 2020.
32. Gschwandtner, M.; Kwitt, R.; Uhl, A.; Pree, W. BlenSor: Blender Sensor Simulation Toolbox. 09 2011, Vol. 6939, pp. 199–208. https://doi.org/10.1007/978-3-642-24031-7_20.
33. Yanovsky, I.; Dragomiretskiy, K. Variational Destriping in Remote Sensing Imagery: Total Variation with L1 Fidelity. *Remote Sensing* **2018**, *10*. https://doi.org/10.3390/rs10020300.
34. Li, C.; Li, Y.; Zhao, H.; Ding, L. Enhancing brain image quality with 3D U-net for stripe removal in light sheet fluorescence microscopy. *Brain Informatics* **2024**, *11*, 24. https://doi.org/10.1186/s40708-024-00236-9.

35. Zhang, K.; Zuo, W.; Chen, Y.; Meng, D.; Zhang, L. Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising. *IEEE Transactions on Image Processing* **2017**, *26*, 3142–3155. https://doi.org/10.1109/TIP.2017.2662206.
36. Kumbar, A.; Anvekar, T.; Vikrama, T.; Tabib, R.; Uma, M. TP-NoDe: Topology-aware Progressive Noising and Denoising of Point Clouds towards Upsampling. 10 2023, pp. 2264–2274. https://doi.org/10.1109/ICCVW60793.2023.00241.
37. Sterzentsenko, V.; Saroglou, L.; Chatzitofis, A.; Thermos, S.; Zioulis, N.; Doumanoglou, A.; Zarpalas, D.; Daras, P. Self-Supervised Deep Depth Denoising **2019**.
38. Shan, W. Optimized Design of 3D Spatial Images Based on Kalman Filter Equation. *Advances in Mathematical Physics* **2021**, *2021*, 1–11. https://doi.org/10.1155/2021/7262871.
39. Zhu, Z.; Chaudhry, E.; Wang, S.; Xia, Y.; Iglesias, A.; You, L.; Zhang, J.J. Shape Reconstruction from Point Clouds Using Closed Form Solution of a Fourth-Order Partial Differential Equation. In Proceedings of the ICCS 2021: International Conference on Computational Science, Krakow, Poland, 2021; Lecture Notes in Computer Science, pp. 207–220. https://doi.org/10.1007/978-3-030-77977-1_16.
40. Rakotosaona, M.J.; Barbera, V.L.; Guerrero, P.; Mitra, N.J.; Ovsjanikov, M. PointCleanNet: Learning to Denoise and Remove Outliers from Dense Point Clouds **2019**.
41. Pistilli, F.; Fracastoro, G.; Valsesia, D.; Magli, E. Learning Robust Graph-Convolutional Representations for Point Cloud Denoising. *IEEE Journal on Selected Topics in Signal Processing* **2021**, *15*, 402–414. https://doi.org/10.1109/JSTSP.2020.3047471.
42. Liu, Z.; Li, Y.; Wang, W.; Liu, L.; Chen, R. Mesh Total Generalized Variation for Denoising **2021**.
43. Bredies, K.; Kunisch, K.; Pock, T. Total Generalized Variation. *SIAM Journal on Imaging Sciences* **2010**, *3*, 492–526. https://doi.org/10.1137/090769521.
44. Liu, Z.; Zhong, S.; Xie, Z.; Wang, W. A novel anisotropic second order regularization for mesh denoising. *Comput. Aided Geom. Des.* **2019**, *71*, 190–201. https://doi.org/10.1016/j.cagd.2019.04.013.
45. Zhang, H.; Wu, C.; Zhang, J.; Deng, J. Variational Mesh Denoising Using Total Variation and Piecewise Constant Function Space. *IEEE Transactions on Visualization and Computer Graphics* **2015**, *21*, 1–1. https://doi.org/10.1109/TVCG.2015.2398432.
46. Li, X.; Zhu, L.; Fu, C.W.; Heng, P.A. Non-Local Low-Rank Normal Filtering for Mesh Denoising. *Computer Graphics Forum* **2018**, *37*, 155–166, [https://onlinelibrary.wiley.com/doi/pdf/10.1111/cgf.13556]. https://doi.org/https://doi.org/10.1111/cgf.13556.
47. He, L.; Schaefer, S. Mesh denoising via L0 minimization. *ACM Trans. Graph.* **2013**, *32*. https://doi.org/10.1145/2461912.2461965.
48. Hurtado, J.; Gattass, M.; Raposo, A. 3D point cloud denoising using anisotropic neighborhoods and a novel sharp feature detection algorithm. *Vis. Comput.* **2022**, *39*, 5823–5848. https://doi.org/10.1007/s00371-022-02698-6.
49. Xu, W.; Dai, W.; Xue, D.; Zheng, Z.; Li, C.; Zou, J.; Xiong, H. Point Cloud Denoising With Fine-Granularity Dynamic Graph Convolutional Networks. *ArXiv* **2024**, *abs/2411.14158*.
50. Hermosilla, P.; Ritschel, T.; Ropinski, T. Total Denoising: Unsupervised Learning of 3D Point Cloud Cleaning, 2019, [arXiv:cs.CV/1904.07615].
51. Xie, H.; McDonnell, K.; Qin, H. Surface reconstruction of noisy and defective data sets. In Proceedings of the IEEE Visualization 2004, 2004, pp. 259–266. https://doi.org/10.1109/VISUAL.2004.101.
52. Kolluri, R.; Shewchuk, J.R.; O'Brien, J.F. Spectral surface reconstruction from noisy point clouds. In Proceedings of the Proceedings of the 2004 Eurographics/ACM SIGGRAPH Symposium on Geometry Processing, New York, NY, USA, 2004; SGP '04, p. 11–21. https://doi.org/10.1145/1057432.1057434.
53. Luo, S.; Hu, W. Differentiable Manifold Reconstruction for Point Cloud Denoising **2020**. https://doi.org/10.1145/3394171.3413727.
54. Serna, A.; Marcotegui, B.; Goulette, F.; Deschaud, J.E. Paris-rue-Madame database: a 3D mobile laser scanner dataset for benchmarking urban detection, segmentation and classification methods. 03 2014, pp. 1–4.
55. Yan, S.; Wu, C.; Wang, L.; Xu, F.; An, L.; Guo, K.; Liu, Y. DDRNet: Depth Map Denoising and Refinement for Consumer Depth Cameras Using Cascaded CNNs. In Proceedings of the Proceedings of the European Conference on Computer Vision (ECCV), September 2018.
56. Zhang, Q.; Shen, X.; Xu, L.; Jia, J. Rolling Guidance Filter. 09 2014, Vol. 8691, pp. 815–830. https://doi.org/10.1007/978-3-319-10578-9_53.
57. Zhang, Z. Image Noise : Detection , Measurement , and Removal Techniques. 2015.

58. Charron, N.; Phillips, S.; Waslander, S.L. De-noising of Lidar Point Clouds Corrupted by Snowfall. *2018 15th Conference on Computer and Robot Vision (CRV)* **2018**, pp. 254–261.
59. Rusu, R.B.; Cousins, S. 3D is here: Point Cloud Library (PCL). In Proceedings of the 2011 IEEE International Conference on Robotics and Automation, 2011, pp. 1–4. https://doi.org/10.1109/ICRA.2011.5980567.
60. Piewak, F.; Pinggera, P.; Schäfer, M.; Peter, D.; Schwarz, B.; Schneider, N.; Pfeiffer, D.; Enzweiler, M.; Zöllner, M. Boosting LiDAR-based Semantic Labeling by Cross-Modal Training Data Generation, 2018, [arXiv:cs.CV/1804.09915].
61. Li, S.; Zhang, B.; Yang, X.; Zhu, W. Edge-guided second-order total generalized variation for Gaussian noise removal from depth map. *Scientific reports* **2020**, *10*, 16329. https://doi.org/10.1038/s41598-020-73342-3.
62. Gu, S.; Zhang, L.; Zuo, W.; Feng, X. Weighted Nuclear Norm Minimization with Application to Image Denoising. In Proceedings of the 2014 IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 2862–2869. https://doi.org/10.1109/CVPR.2014.366.
63. Liu, Y.; Zou, B.; Xu, J.; Yang, S.; Li, Y. Denoising for 3D Point Cloud Based on Regularization of a Statistical Low-Dimensional Manifold. *Sensors* **2022**, *22*. https://doi.org/10.3390/s22072666.
64. Cortinhal, T.; Tzelepis, G.; Aksoy, E.E. SalsaNext: Fast Semantic Segmentation of LiDAR Point Clouds for Autonomous Driving. *CoRR* **2020**, *abs/2003.03653*, [2003.03653].
65. Zhu, X.; Zhou, H.; Wang, T.; Hong, F.; Ma, Y.; Li, W.; Li, H.; Lin, D. Cylindrical and Asymmetrical 3D Convolution Networks for LiDAR Segmentation, 2020, [arXiv:cs.CV/2011.10033].
66. Lai, X.; Chen, Y.; Lu, F.; Liu, J.; Jia, J. Spherical Transformer for LiDAR-based 3D Recognition, 2023, [arXiv:cs.CV/2303.12766].
67. Seppanen, A.; Ojala, R.; Tammi, K. 4DenoiseNet: Adverse Weather Denoising From Adjacent Point Clouds. *IEEE Robotics and Automation Letters* **2023**, *8*, 456–463. https://doi.org/10.1109/lra.2022.3227863.
68. Yu, M.Y.; Vasudevan, R.; Johnson-Roberson, M. LiSnowNet: Real-time Snow Removal for LiDAR Point Cloud, 2022, [arXiv:cs.CV/2211.10023].
69. Raisuddin, A.M.; Cortinhal, T.; Holmblad, J.; Aksoy, E.E. 3D-OutDet: A Fast and Memory Efficient Outlier Detector for 3D LiDAR Point Clouds in Adverse Weather. In Proceedings of the 2024 IEEE Intelligent Vehicles Symposium (IV), 2024, pp. 2862–2868. https://doi.org/10.1109/IV55156.2024.10588582.
70. Park, J.I.; Park, J.; Kim, K.S. Fast and Accurate Desnowing Algorithm for LiDAR Point Clouds. *IEEE Access* **2020**, *8*, 160202–160212. https://doi.org/10.1109/ACCESS.2020.3020266.
71. Wang, W.; You, X.; Chen, L.; Tian, J.; Tang, F.; Zhang, L. A Scalable and Accurate De-Snowing Algorithm for LiDAR Point Clouds in Winter. *Remote Sensing* **2022**, *14*. https://doi.org/10.3390/rs14061468.
72. Wang, W.; Yang, T.; Du, Y.; Liu, Y. Snow Removal for LiDAR Point Clouds With Spatio-Temporal Conditional Random Fields. *IEEE Robotics and Automation Letters* **2023**, *8*, 6739–6746. https://doi.org/10.1109/LRA.2023.3311360.
73. Kurup, A.; Bos, J. DSOR: A Scalable Statistical Filter for Removing Falling Snow from LiDAR Point Clouds in Severe Winter Weather, 2021, [arXiv:cs.CV/2109.07078].
74. Mao, A.; Du, Z.; Wen, Y.H.; Xuan, J.; Liu, Y.J. PD-Flow: A Point Cloud Denoising Framework with Normalizing Flows **2022**.
75. Xie, Y.; Yuan, M.; Dong, B.; Li, Q. Unsupervised Image Denoising with Score Function, 2023, [arXiv:cs.CV/2304.08384].
76. Kim, K.; Ye, J.C. Noise2Score: Tweedie's Approach to Self-Supervised Image Denoising without Clean Images, 2021, [arXiv:eess.IV/2106.07009].
77. Huang, T.; Li, S.; Jia, X.; Lu, H.; Liu, J. Neighbor2Neighbor: Self-Supervised Denoising from Single Noisy Images, 2021, [arXiv:eess.IV/2101.02824].

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.